

Differentially Private Formation Control: Privacy and Network Co-Design*

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Abstract

Privacy in multi-agent control is receiving increased attention, though often a networked system and privacy protections are designed separately, which can harm performance. Therefore, this paper presents a co-design framework for networks and private controllers, and we apply it to private formation control. Agents' state trajectories are protected using differential privacy, and we quantify its impact by bounding the steady-state error for private formations. Then, we analyze tradeoffs between privacy level, system performance, and connectedness of the network's communication topology. These tradeoffs are used to formulate a co-design optimization framework to jointly design agents' communication topology and their privacy levels. Simulation results illustrate the success of this framework.

Keywords: Network analysis and control, Privacy, Optimization

1 Introduction

Multi-agent systems require agents to share information to collaborate, and in some cases this information is sensitive. For example, self-driving cars may share location data to be routed to a destination. Location data and other streams can be revealing and thus require protections, though this data must still be useful for multi-agent coordination.

This type of privacy has recently been achieved using differential privacy [1], which has been applied to various control, estimation, and optimization problems [2, 3, 4, 5, 6, 7, 8]. Broadly speaking, differential privacy carefully introduces randomness in a system, e.g., by adding noise to conceal sensitive information. Differential privacy is appealing because it is immune to post-processing and robust to side information [9]. These properties mean that privacy guarantees are not compromised by performing operations on differentially private data, and that they are not weakened by much by an adversary with additional information about data-producing agents [10].

In this work we focus on distributed formation control, e.g., robots physically assembling into geometric shapes or non-physical agents maintaining relative state offsets. We develop a private decentralized formation control protocol in which each agent adds noise to keep its state trajectory private from other agents. We provide privacy guarantees over an agent's entire state trajectory, which ensures that an adversary is unlikely to learn much about the state trajectory of a system. Privacy noise of course impacts agents' formations, and we provide a closed-form for the steady-state error covariance that privacy induces in formations. We also bound the steady-state error itself, and both bounds are in terms of agents' privacy parameters.

Privacy is often only a post-hoc concern that is incorporated after a network and/or a controller is designed, which can harm performance; however, there also exist approaches that achieve privacy without sacrificing convergence [11]. Therefore, we leverage our steady-state analyses to formulate optimization problems that design a network topology and a differential privacy implementation together.

While other works have implemented other forms of privacy for trajectories in consensus problems, e.g., [12], to the best of our knowledge, we are the first to implement trajectory-level *differential* privacy in the setting of formation control. The formation control protocol does resemble the average consensus

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差异化隐私形成控制：隐私与网络协同设计

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抽象的

多智能体控制中的隐私问题日益受到关注，但通常情况下，网络系统和隐私保护机制是分开设计的，这可能会影响系统性能。因此，本文提出了一种网络与私有控制器协同设计的框架，并将其应用于私有编队控制。我们使用差分隐私来保护智能体的状态轨迹，并通过限制私有编队的稳态误差来量化其影响。然后，我们分析了隐私级别、系统性能和网络通信拓扑连通性之间的权衡关系。这些权衡关系被用于构建一个协同设计优化框架，以联合设计智能体的通信拓扑结构及其隐私级别。仿真结果验证了该框架的有效性。

关键词：网络分析与控制、隐私、优化

1 介绍

多智能体系统要求智能体之间共享信息以进行协作，而某些情况下这些信息较为敏感。例如，自动驾驶汽车可能需要共享位置数据以便规划路线。位置数据和其他信息流可能泄露敏感信息，因此需要采取保护措施，但这些数据对于多智能体协调仍然至关重要。

近年来，差分隐私[1]已成为实现此类隐私保护的有效方法，并已应用于各种控制、估计和优化问题[2, 3, 4, 5, 6, 7, 8]。简而言之，差分隐私通过在系统中引入随机性（例如，通过添加噪声来隐藏敏感信息）来达到目的。差分隐私的优势在于它不受后处理的影响，并且对辅助信息具有鲁棒性[9]。这些特性意味着，对差分隐私数据进行操作不会损害隐私保证，即使攻击者掌握了关于数据生成主体的额外信息，也不会显著削弱隐私保证[10]。

本文重点研究分布式编队控制，例如机器人物理组装成几何形状或非物理智能体维持相对状态偏移。我们开发了一种私有的去中心化编队控制协议，其中每个智能体通过添加噪声来保护其状态轨迹，使其不被其他智能体获取。我们提供了智能体整个状态轨迹的隐私保证，从而确保攻击者难以获取系统状态轨迹的大量信息。隐私噪声自然会影晌智能体的编队，我们给出了隐私噪声在编队中引起的稳态误差协方差的闭式解。此外，我们还对稳态误差本身进行了界定，这两个界定均与智能体的隐私参数有关。

隐私通常只是在网络和/或控制器设计完成后才考虑的事后因素，这可能会损害性能；然而，也存在一些方法可以在不牺牲收敛性的前提下实现隐私保护[11]。因此，我们利用稳态分析来构建优化问题，从而同时设计网络拓扑和差分隐私实现。

虽然其他工作已在共识问题中实现了轨迹隐私保护的其他形式，例如[12]，但据我们所知，我们是第一个在编队控制场景下实现轨迹级差分隐私的。编队控制协议与平均共识协议类似。

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protocol, and existing work has applied differential privacy to initial states in consensus, e.g., [13, 14, 15] and others. This paper differs by implementing trajectory-level differential privacy [16], which conceals deviations from a nominal state trajectory, which can occur at any time and can occur across multiple timesteps, not just the initial time. There also exists work that considers differential privacy and *co-design* in a control setting [17], which considers initial state privacy and uses co-design to tune privacy noise and controller parameters, not the network topology. There also exists work on the co-design of control and communication, without addressing privacy [18].

This difference in privacy is necessary because we address a fundamentally different type of privacy need than initial-state privacy. To highlight the difference, suppose that a coalition of drones is assembling a formation, and, in the course of doing so, one drone takes a detour to photograph a location of interest, which it would like to conceal. This detour occurs after the initial time, and it cannot be concealed by privatizing its initial state. However, such a detour can be concealed by implementing privacy at the trajectory level, which is what we do.

Beyond differential privacy, other privacy-preserving approaches have been applied to networked control and estimation. For example, cryptographic approaches such as homomorphic encryption enable computation over a secure, shared channel at the cost of increased computational overhead [19]. Estimation-theoretic approaches based on information bounds have also been used to limit information disclosure by constraining inference accuracy [20]. In this work, we focus on differential privacy because it provides a formal, worst-case notion of privacy that is independent of adversary models and is well suited to decentralized control architectures.

While the privacy mechanism and some optimization formalisms employed in this work are standard, the problem formulation and analysis that connect privacy, network topology, and formation control performance are not. Our contributions therefore lie in making these connections analytically and developing a computational framework to apply them in practice.

In detail, the main contributions of this paper are as follows:

- An analytical characterization of the steady-state formation error under trajectory-level differential privacy as a function of the network topology
- A co-design formulation that jointly selects agent-level privacy parameters and communication weights under steady-state performance constraints
- A tractable solution approach based on a biconvex relaxation and alternate convex search which enables practical computation of co-designed networks.

The main analytical challenge in this work is to bound the steady-state formation error under trajectory-level differential privacy in a manner that explicitly captures the joint dependence on agent-level privacy parameters and the communication topology. This dependence is nonlinear and coupled, and therefore difficult to optimize directly. To overcome this difficulty we derive an explicit upper bound on steady-state error that isolates the dependence on privacy parameters and algebraic connectivity. The resulting bound is structured so that it can be embedded directly into a co-design optimization problem, enabling principled trade-offs between privacy strength, network connectivity, and formation performance.

A preliminary version of this work appeared in [21]. This paper’s new contributions are the co-design framework, the steady-state error analysis, new simulations, and proofs of all results. The rest of the paper is organized as follows. Section 2 gives background, and Section 3 provides problem statements. Then Section 4 analyzes private formation control, and Section 5 develops the co-design framework. Section 6 provides simulations, and Section 7 concludes.

Notation: $\mathbb{R}$ denotes the reals and $\mathbb{N}$ denotes the naturals. For $\ell \in \mathbb{N}$, let $[\ell] := \{1, \dots, \ell\}$. We use $I_a \in \mathbb{R}^{a \times a}$ for the identity matrix in a dimensions and $\mathbf{1}$ for the vector of all ones in $\mathbb{R}^N$. We use the norm $\|v\|_\infty = \max_{i \in [N]} |v_i|$.

2 Background and Preliminaries

Here we review graph theory and differential privacy.

协议方面，现有工作已将差分隐私应用于共识中的初始状态，例如[13, 14, 15]等。本文的不同之处在于实现了轨迹级差分隐私[16]，它能够隐藏偏离标称状态轨迹的情况，这种偏差可能发生在任何时刻，并且跨越多个时间步，而不仅仅是初始时刻。此外，还有一些工作在控制设置中考虑了差分隐私和协同设计[17]，这些工作考虑了初始状态隐私，并使用协同设计来调整隐私噪声和控制器参数，而不是网络拓扑结构。还有一些工作研究了控制和通信的协同设计，但没有涉及隐私问题[18]。

这种隐私差异是必要的，因为我们处理的隐私需求与初始状态隐私有着本质的不同。为了突出这种差异，假设一群无人机正在集结编队，在此过程中，其中一架无人机绕道拍摄一个它想要隐藏的目标位置。这次绕道发生在初始状态之后，因此无法通过对其初始状态进行隐私保护来隐藏。然而，通过在轨迹层面实施隐私保护，就可以隐藏这种绕道行为，而这正是我们所做的。

除了差分隐私之外，其他隐私保护方法也被应用于网络控制和估计。例如，同态加密等密码学方法允许在安全的共享信道上进行计算，但代价是计算开销增加[19]。基于信息界限的估计理论方法也被用于通过限制推理精度来限制信息泄露[20]。本文重点关注差分隐私，因为它提供了一种形式化的、最坏情况下的隐私概念，该概念独立于攻击者模型，并且非常适合去中心化控制架构。

虽然本文采用的隐私机制和一些优化方法都是标准的，但将隐私、网络拓扑和编队控制性能联系起来的问题描述和分析却并非如此。因此，我们的贡献在于从理论上建立这些联系，并开发一个计算框架以将其应用于实践。

具体而言，本文的主要贡献如下：

- 对轨迹级差分隐私下稳态形成误差进行分析刻画，并将其作为网络拓扑的函数
- 一种协同设计方案，可在稳态性能约束下联合选择代理级隐私参数和通信权重。
- 一种基于双凸松弛和交替凸搜索的易于处理的解决方案方法，能够对协同设计的网络进行实际计算。

本文的主要分析挑战在于，如何在轨迹级差分隐私的框架下，明确地捕捉到对代理级隐私参数和通信拓扑结构的联合依赖性，从而界定稳态编队误差。这种依赖性是非线性和耦合的，因此难以直接优化。为了克服这一困难，我们推导出了稳态误差的显式上界，该上界分离了对隐私参数和代数连通性的依赖性。所得上界的结构使其可以直接嵌入到协同设计优化问题中，从而能够在隐私强度、网络连通性和编队性能之间进行合理的权衡。

本文的初步版本发表于[21]。本文的新贡献包括协同设计框架、稳态误差分析、新的仿真以及所有结果的证明。本文的其余部分组织如下：第2节介绍背景，第3节阐述问题，第4节分析私有编队控制，第5节构建协同设计框架，第6节提供仿真结果，第7节总结全文。

符号说明： $\mathbb{R}$ 表示实数集， $\mathbb{N}$ 表示自然数集。对于 $\ell \in \mathbb{N}$ ，令 $[\ell] := \{1, \dots, \ell\}$ 。我们用 $I \in \mathbb{R}$ 表示 n 维单位矩阵，用 $\mathbf{1}$ 表示 $\mathbb{R}$ 中全为 1 的向量。我们使用范数 $\|v\| = \max|v|$ 。

2 背景和预备知识

本文将回顾图论和差分隐私。

2.1 Graph Theory Background

A graph $\mathcal{G} = (V, E)$ is defined over a set of N nodes indexed over $V = \{1, \dots, N\}$ and a set of edges $E \subseteq V \times V$. The pair $(i, j) \in E$ if nodes i and j share a connection and $(i, j) \notin E$ if they do not. This paper considers undirected, weighted, simple graphs. Undirectedness means that an edge $(i, j) \in E$ is not distinguished from $(j, i) \in E$. Simplicity means that $(i, i) \notin E$ for all $i \in V$. Weightedness means that the edge $(i, j) \in E$ has a weight $w_{ij} = w_{ji} > 0$.

Definition 1 (Connected Graph). *A graph $\mathcal{G}$ is connected if, for all $i, j \in \{1, \dots, N\}$, $i \neq j$, there is a sequence of edges one can traverse from node i to node j .*

We assume the following in the rest of the paper.

Assumption 1. *Agents communicate over a connected, weighted, simple, undirected graph $\mathcal{G} = (V, E)$ on N nodes.*

This paper uses the weighted graph Laplacian, which is defined with weighted adjacency and degree matrices. The weighted adjacency matrix $A(\mathcal{G}) \in \mathbb{R}^{N \times N}$ of $\mathcal{G}$ is given by

$$A(\mathcal{G})_{ij} = \begin{cases} w_{ij} & (i, j) \in E \\ 0 & \text{otherwise} \end{cases}.$$

For undirected $\mathcal{G}$, $A(\mathcal{G})$ is symmetric. The degree matrix is $D(\mathcal{G}) = \text{diag}(d_1, \dots, d_N)$, and the Laplacian is $L(\mathcal{G}) = D(\mathcal{G}) - A(\mathcal{G})$. Agent i 's neighborhood set N_i is the set of all agents that agent i communicates with, i.e., $N_i = \{j \mid (i, j) \in E\}$. The weighted degree of node i is $d_i = \sum_{j \in N_i} w_{ij}$.

Let $\lambda_k(\cdot)$ be the k^{th} smallest eigenvalue of a symmetric matrix. Then for all graphs $0 = \lambda_1(L(\mathcal{G})) \leq \lambda_2(L(\mathcal{G})) \leq \dots \leq \lambda_N(L(\mathcal{G}))$.

Definition 2 (Algebraic Connectivity [22]). *The algebraic connectivity of a graph $\mathcal{G}$ is the second smallest eigenvalue of $L(\mathcal{G})$, and $\mathcal{G}$ is connected if and only if $\lambda_2(L(\mathcal{G})) > 0$.*

2.2 Differential Privacy Background

This section briefly reviews differential privacy. More detail can be found in [16, 23]. The goal of differential privacy is to make similar pieces of data appear approximately indistinguishable, and it is immune to post-processing [23]. That is, arbitrary post-hoc computations on private data do not harm differential privacy, including filtering [16, 24].

We apply differential privacy to agents' state trajectories, and we consider vector-valued trajectories of the form $Z = (Z(1), Z(2), \dots, Z(k), \dots)$, where $Z(k) \in \mathbb{R}^d$ for all k . The ℓ_2 norm of Z is defined as $\|Z\|_{\ell_2} = (\sum_{k=1}^{\infty} \|Z(k)\|_2^2)^{\frac{1}{2}}$, where $\|\cdot\|_2$ is the ordinary 2-norm on $\mathbb{R}^d$. We privatize trajectories in $\ell_{\infty}^d = \{Z \mid \|Z(k)\|_{\infty} < \infty \text{ for all } k\}$, namely the set of vector-valued trajectories whose entries are all finite. The set ℓ_{∞}^d contains trajectories that do not converge, which admits a wide variety of trajectories seen in control systems.

We consider a network of N agents, where agent i 's state trajectory is denoted by x_i . Its k^{th} element is denoted $x_i(k) \in \mathbb{R}^d$ for $d \in \mathbb{N}$, and agent i 's state trajectory belongs to ℓ_{∞}^d . In this work, agents privatize their state trajectories before they are shared, which contrasts with works that privatize collections of trajectories. The approach we take is called *input perturbation* [16], and it amounts to privatizing data, then using it in some computation. This is in place of *output perturbation*, in which one performs some computation with sensitive data and then privatizes its output. In principle, one may envision using output perturbation to privatize agents' state trajectories. However, the use of output perturbation requires aggregating agents' states in one place, and such an aggregator is typically absent in formation control.

Indeed, if an aggregator were available, then all agents could send their initial states to that aggregator, it could execute the formation control protocol in a centralized way, and then the aggregator could send each agent their final state to travel to. The decentralized formation control protocol is thus only meaningful in the absence of an aggregator, and we therefore choose a privacy implementation that fits with the lack of an aggregator, which is input perturbation.

2.1 图论背景

图 $G = (V, E)$ 定义在包含 N 个节点的集合 $V = \{1, \dots, N\}$ 和边的集合 $E \subseteq V \times V$ 上。如果节点 i 和 j 之间存在连接，则 $(i, j) \in E$ ；否则， $(i, j) \notin E$ 。本文考虑无向、加权、简单图。无向性是指边 $(i, j) \in E$ 与 $(j, i) \in E$ 之间没有区别。简单性是指对于所有 $i \in V$ ， $(i, i) \notin E$ 。加权性是指边 $(i, j) \in E$ 具有权重 w ，且 $w > 0$ 。

定义 1（连通图）。如果对于所有 $i, j \in \{1, \dots, N\}$ ， $i \neq j$ ，存在一条边序列可以从节点 i 遍历到节点 j ，则称图 G 是连通的。

本文余下部分将作如下假设。

假设 1. 代理通过 N 个节点的连通加权简单无向图 $G = (V, E)$ 进行通信。

本文使用加权图拉普拉斯算子，它由加权邻接矩阵和加权重矩阵定义。加权邻接矩阵 $A(G) \in \mathbb{R}$ 表示图 G ，其表达式为：

$$A(G) = \begin{pmatrix} w(i, j) & \text{if } (i, j) \in E \\ 0 & \text{otherwise} \end{pmatrix}.$$

对于无向图 G ， $A(G)$ 是对称的。度矩阵为 $D(G) = \text{diag}(d, \dots, d)$ ，拉普拉斯矩阵为 $L(G) = D(G) - A(G)$ 。智能体 i 的邻域集 N 是与智能体 i 通信的所有智能体的集合，即 $N = \{j \mid (i, j) \in E\}$ 。节点 i 的加权重为 $d = \sum_{j \in N} w$ 。

设 $\lambda(\cdot)$ 为对称矩阵的 k 个最小特征值。则对于所有图， $0 = \lambda_1(L(G)) \leq \lambda_2(L(G)) \leq \dots \leq \lambda_N(L(G))$ 。

定义 2（代数连通性 [22]）。图 G 的代数连通性是 $L(G)$ 的第二小特征值， G 连通当且仅当 $\lambda_2(L(G)) > 0$ 。

2.2 差分隐私背景

本节简要回顾差分隐私。更多细节可参见[16, 23]。差分隐私的目标是使相似的数据片段看起来几乎无法区分，并且它不受后处理的影响[23]。也就是说，对私有数据进行任意的后处理计算，包括滤波[16, 24]，都不会损害差分隐私。

我们将差分隐私应用于智能体的状态轨迹，并考虑形如 $Z = (Z(1), Z(2), \dots, Z(k), \dots)$ 的向量值轨迹，其中 $Z(k) \in \mathbb{R}^d$ 对所有 k 成立。 Z 的 ℓ_2 范数定义为

其中 $\|\cdot\|_2$ 是 $\mathbb{R}^d$ 上的普通 2-范数。我们将轨迹私有化到 $\ell = \{Z \mid \|Z(k)\|_2 < \infty \text{ 对所有 } k \text{ 成立}\}$ ，即所有元素均为有限的向量值轨迹的集合。集合 ℓ 包含不收敛的轨迹，这涵盖了控制系统中常见的各种轨迹。

我们考虑一个由 N 个智能体组成的网络，其中智能体 i 的状态轨迹用 x 表示。其元素记为 $x(k) \in \mathbb{R}^d$ ，其中 $d \in \mathbb{N}$ ，且智能体 i 的状态轨迹属于 ℓ 。本文中，智能体在共享状态轨迹之前对其进行私有化，这与对轨迹集合进行私有化的工作形成对比。我们采用的方法称为输入扰动 [16]，它相当于对数据进行私有化，然后将其用于某些计算。这与输出扰动不同，输出扰动是指对敏感数据进行某些计算，然后对其输出进行私有化。原则上，可以设想使用输出扰动来对智能体的状态轨迹进行私有化。然而，使用输出扰动需要将智能体的状态聚合到一个地方，而这种聚合器在编队控制中通常是不存在的。

事实上，如果存在聚合器，所有智能体都可以将其初始状态发送给该聚合器，聚合器可以集中执行编队控制协议，然后聚合器可以将每个智能体最终到达的状态发送给它们。因此，去中心化的编队控制协议只有在没有聚合器的情况下才有意义，所以我们选择了一种适合缺少聚合器的隐私实现方式，即输入扰动。

This setup has the advantages of allowing each agent to calibrate its own privacy parameters, and it also ensures that agents' information is privatized before it is ever shared. A known challenge can be that more noise is often required for input perturbation privacy than in alternative setups.

The goal of differential privacy is to make “similar” pieces of data approximately indistinguishable, and an adjacency relation quantifies when pieces of data are “similar.” For input perturbation, adjacency is defined for single agents.

Definition 3 (Adjacency). *Fix an adjacency parameter $b_i > 0$ for agent i . Then $\text{Adj}_{b_i} : \ell_\infty^d \times \ell_\infty^d \rightarrow \{0, 1\}$ is defined as*

$$\text{Adj}_{b_i}(v_i, r_i) = \begin{cases} 1 & \|v_i - r_i\|_{\ell_2} \leq b_i \\ 0 & \text{otherwise.} \end{cases}$$

In words, two state trajectories that agent i could produce are adjacent if and only if the ℓ_2 -norm of their difference is upper bounded by b_i . This means that every state trajectory within ℓ_2 distance b_i from agent i 's actual state trajectory must be made approximately indistinguishable from it. For example, agent i may take a detour of size b_i from the nominal trajectory that it would follow to get into formation. This detour can occur at any time, and the adjacency relation in Definition 3 requires that our privacy implementation conceal it, regardless of when it occurs. Existing approaches to initial-state privacy do not provide these protections.

While adjacency requires two state trajectories to have a finite difference between them, it does not require the two trajectories in question to converge. This is reflected in the fact that an agent's trajectories are contained in the space ℓ_∞^d , which indicates that trajectories consist of real-valued vectors in $\mathbb{R}^d$ that can take any finite values, and, in particular, they need not ever converge to any value.

Privacy is implemented by a “mechanism,” which is a randomized map that we define next. First, fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We consider outputs in ℓ_∞^d and use a σ -algebra over ℓ_∞^d , denoted $\mathcal{F}_\infty^d$ [25].

Definition 4 (Differential Privacy). *Let $\epsilon_i > 0$ and $\delta_i \in (0, \frac{1}{2})$ be given. Fix $b_i > 0$. A mechanism $M : \ell_\infty^d \times \Omega \rightarrow \ell_\infty^d$ is (ϵ_i, δ_i) -differentially private if, for all adjacent $x_i, x'_i \in \ell_\infty^d$, we have*

$$\mathbb{P}[M(x_i) \in S] \leq e^{\epsilon_i} \mathbb{P}[M(x'_i) \in S] + \delta_i$$

for all $S \in \mathcal{F}_\infty^d$.

Definition 4 is applied individually to each agent $i \in [N]$, i.e., each agent's released trajectory satisfies (ϵ_i, δ_i) -differential privacy with respect to Definition 3. To calibrate privacy, agent i selects privacy parameters ϵ_i and δ_i . Typically, $\epsilon_i \in [0.1, \ln 3]$ and $\delta_i \in [0, 0.05]$ for all i [16, 26, 10]. The value of δ_i can be regarded as allowing a small probability that the ϵ_i -differential privacy guarantee for agent i is violated, while ϵ_i controls the strength of privacy for agent i [9, Section 2.3].

In this work, we use the Gaussian mechanism, which adds zero-mean i.i.d. noise drawn from a Gaussian distribution. We define it using the Q -function, i.e., $Q(y) = \frac{1}{\sqrt{2\pi}} \int_y^\infty e^{-\frac{z^2}{2}} dz$.

Lemma 1 (Gaussian Mechanism [16]). *Let $b_i > 0$, $\epsilon_i > 0$, and $\delta_i \in (0, \frac{1}{2})$ be given, and let $x_i \in \ell_\infty^d$. Then the Gaussian mechanism takes the form $\tilde{x}_i(k) = x_i(k) + v_i(k)$, where v_i is a stochastic process with $v_i(k) \sim \mathcal{N}(0, \Xi_{v_i})$, where $\Xi_{v_i} = \sigma_i^2 I_d$ with $\sigma_i \geq \frac{b_i}{2\epsilon_i} (K_{\delta_i} + \sqrt{K_{\delta_i}^2 + 2\epsilon_i})$ and $K_{\delta_i} = Q^{-1}(\delta_i)$. This mechanism provides (ϵ_i, δ_i) -differential privacy to x_i .*

3 Problem Formulation

3.1 Problem Statement

Problem 1. *Let Assumption 1 hold. Let $x_i(k) \in \mathbb{R}^d$ be agent i 's state at time k , N_i be agent i 's neighborhood set, and $\gamma > 0$ be a stepsize. Let $n_i(k)$ denote the process noise in agent i 's state dynamics at time k . We define $\Delta_{ij} \in \mathbb{R}^d$ for all $(i, j) \in E$ as the desired relative state offset between agents i and j . Do each of the following:*

[i.]

这种设置的优势在于允许每个代理自行校准隐私参数，并且还能确保代理的信息在共享之前就被保密。但一个已知的挑战是，与其它设置相比，这种设置通常需要更多的噪声才能实现输入扰动隐私。

差分隐私的目标是使“相似”的数据片段近似无法区分，而邻接关系则量化了数据片段何时“相似”。对于输入扰动，邻接关系是针对单个代理定义的。

定义 3 (邻接关系)。对于智能体 i ，固定邻接参数 $b > 0$ 。则 $\text{Adj}: \ell \times \ell \rightarrow \{0, 1\}$ 定义为：

$$\text{Adj}(v, r) = \begin{cases} 1 & \|v - r\| \leq b \\ 0 & \text{否则。} \end{cases}$$

换句话说，智能体 i 可能产生的两条状态轨迹相邻当且仅当它们之差的 ℓ 范数上界为 b 。这意味着，与智能体 i 的实际状态轨迹相距 ℓ 距离 b 内的任何状态轨迹都必须使其与实际状态轨迹近似无法区分。例如，智能体 i 可能偏离其进入编队时所遵循的标称轨迹一段距离为 b 的路径。这种绕行可能发生在任何时刻，而定义 3 中的邻接关系要求我们的隐私实现必须隐藏这种绕行，无论它何时发生。现有的初始状态隐私方法无法提供这些保护。

邻接性要求两条状态轨迹之间的差值有限，但并不要求这两条轨迹收敛。这体现在智能体的轨迹包含在空间 ℓ 中，这意味着轨迹由 R 中的实值向量构成，这些向量可以取任意有限值，尤其重要的是，它们无需收敛到任何特定值。

隐私是通过一种“机制”来实现的，该机制是一个随机映射，我们将在下文中定义。首先，固定一个概率空间 (Ω, F, P) 。我们考虑 ℓ 中的输出，并使用 ℓ 上的 σ 代数，记为 $F[25]$ 。

定义 4 (差分隐私)。设 $\epsilon > 0$ 且 $\delta \in (0, 1)$ 。固定 $b > 0$ 。机制 $M: \ell \times \Omega \rightarrow \ell$ 是 (ϵ, δ) -差分隐私的，如果对于所有相邻的 $x, x' \in \ell$ ，我们有

$$P[M(x) \in S] \leq e P[M(x') \in S] + \delta$$

对于所有 $S \in F$ 。

定义 4 分别应用于每个智能体 $i \in [N]$ ，即每个智能体发布的轨迹均满足定义 3 中的 (ϵ, δ) 差分隐私。为了校准隐私，智能体 i 选择隐私参数 ϵ 和 δ 。通常，对于所有 i ， $\epsilon \in [0.1, \ln 3]$ 且 $\delta \in [0, 0.05]$ [16, 26, 10]。 δ 的值可以被视为允许智能体 i 的 ϵ 差分隐私保证被违反的概率较小，而 ϵ 则控制智能体 i 的隐私强度 [9, 第 2.3 节]。

在这项工作中，我们使用高斯机制，它添加了均值为零、服从高斯分布的独立同分布噪声。我们使用 Q 函数来定义它，即 $Q(y) =$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^y e^{-\frac{z^2}{2}} dz。$$

引理 1 (高斯机制 [16])。设 $b > 0$ ， $\epsilon > 0$ ， $\delta \in (0, 1)$ ，且 $x \in \ell$ 。则高斯机制可表示为 $\tilde{x}(k) = x(k) + v(k)$ ，其中 $v(k)$ 为随机过程，且 $v(k) \sim N(0, \Xi)$ ，其中 $\Xi = \sigma I$ ， $\sigma \geq (K + \frac{q}{K + 2\epsilon})$ 和 $K = Q(\delta)$ 。这种机制

为 x 提供 (ϵ, δ) -差分隐私。

3 问题描述

3.1 问题陈述

问题 1. 假设 1 成立。令 $x(k) \in R$ 表示智能体 i 在时刻 k 的状态， N 表示智能体 i 的邻域集合， $\gamma > 0$ 为步长。令 $n(k)$ 表示智能体 i 在时刻 k 的状态动态过程中的噪声。对于所有 $(i, j) \in E$ ，我们定义 $\Delta \in R$ 为智能体 i 和 j 之间期望的相对状态偏移量。完成以下各项：

[我。]

1. Implement the formation control protocol

$$x_i(k+1) = x_i(k) + \gamma \sum_{j \in N_i} w_{ij}(x_j(k) - x_i(k) - \Delta_{ij}) + n_i(k), \quad (1)$$

in a differentially private, decentralized manner.

2. Bound the performance of the network in terms of privacy parameters and network properties.

3. Use those bounds to co-design the communication topology and privacy parameters of the network.

Next, we define agent- and network-level dynamics and detail how each agent will enforce differential privacy.

3.2 Multi-Agent Formation Control

The goal of formation control is for agents to reach a set of relative states. Multi-agent formation control is a well-researched problem and there are several mathematical formulations one can use to achieve similar results [27, 28, 29, 30, 31, 32, 33]. We will define offsets between agents that communicate and the objective is for all agents to maintain these relative offsets.

Assumption 2 (Section 6.1.1 of [33]). *The formation specified by $\{\Delta_{ij}\}_{(i,j) \in E}$ is feasible, i.e., there exists a set of points that simultaneously satisfies all pairwise distances. We require $\Delta_{ij} = -\Delta_{ji}$ for all $(i,j) \in E$.*

The objective is having $\lim_{k \rightarrow \infty} (x_j(k) - x_i(k)) = \Delta_{ij}$ for all $(i,j) \in E$. The formation can be centered on any point in $\mathbb{R}^d$, i.e., we allow formations to be translationally invariant [33].

Remark 1. *For all $i \in [N]$, agent i will execute a privatized form of (1) to be detailed below. To ease discussion, we will consider a set of points $\mathcal{P} = \{p_1, \dots, p_N\}$ below that satisfy $p_j - p_i = \Delta_{ij}$ for all $(i,j) \in E$. We emphasize here that agents do not know (and do not need to know) the point p_i for any $i \in [N]$. Instead, our use of $\mathcal{P}$ is only for ease of discussion and analysis. Algorithm 1 below shows the steps that each agent will actually implement, and none of them require p_i .*

As in (1), we model agents as discrete-time single integrators, i.e., $x_i(k+1) = x_i(k) + u_i(k) + n_i(k)$, where $n_i(k) \sim \mathcal{N}(0, s_i^2 I_d)$ is process noise and $u_i(k) \in \mathbb{R}^d$ is agent i 's input. Let the vector $p = (p_1^T, \dots, p_N^T)^T \in \mathbb{R}^{Nd}$ be a network-level state that satisfies the formation specification, let $x(k) = (x_1(k)^T, \dots, x_N(k)^T)^T \in \mathbb{R}^{Nd}$, and let $n(k) = (n_1(k)^T, \dots, n_N(k)^T)^T \in \mathbb{R}^{Nd}$. Set $\bar{x}(k) = x(k) - p$. Then analyzing (1) at the network-level is equivalent to analyzing

$$\bar{x}(k+1) = \left((I_N - \gamma L(\mathcal{G})) \otimes I_d \right) \bar{x}(k) + n(k), \quad (2)$$

where $\otimes$ is the Kronecker product; we reiterate that (2) is used only for analysis, and agents do not need to know p_i for any i .

Let $\bar{x}_{i[l]}$ be the l^{th} scalar element of $\bar{x}_i$ for $l \in [d]$. Then $\bar{x}_{[l]} = [\bar{x}_{1[l]}, \dots, \bar{x}_{N[l]}]^T$ is the vector of all agents' states in the l^{th} dimension, and $\bar{n}_{[l]} = [n_{1[l]}, \dots, n_{N[l]}]^T$ is the vector of corresponding noise terms. Analyzing (2) is therefore equivalent to analyzing the protocol

$$\bar{x}_{[l]}(k+1) = (I_N - \gamma L(\mathcal{G})) \bar{x}_{[l]}(k) + \bar{n}_{[l]}(k) \quad (3)$$

for all $l \in [d]$ simultaneously.

3.3 Private Communications (Solution to Problem 1.1)

To privately implement the protocol in (1), agent j starts by selecting privacy parameters $\epsilon_j > 0$, $\delta_j \in (0, \frac{1}{2})$, and adjacency relation Adj_{b_j} with $b_j > 0$. Then, agent j privatizes its state trajectory x_j with the Gaussian mechanism. Let $\tilde{x}_j$ denote the differentially private version of x_j , where $\tilde{x}_j(k) = x_j(k) + v_j(k)$, with $v_j(k) \sim \mathcal{N}(0, \Xi_{v_j})$. Here, $\Xi_{v_j} = \sigma_j^2 I_d$ and $\sigma_j \geq \frac{b_j}{2\epsilon_j} (K_{\delta_j} + \sqrt{K_{\delta_j}^2 + 2\epsilon_j})$. Lemma 1 shows that this setup keeps agent j 's state trajectory (ϵ_j, δ_j) -differentially private. Agent j then shares $\tilde{x}_j(k)$ with its neighbors.

1. 实现编队控制协议 $x(k+1) = x(k) + \gamma$

$$\sum_{j \in N} w(x(k) - x(k) - \Delta) + n(k), \quad (1)$$

以差异化隐私、去中心化的方式。

2. 根据隐私参数和网络属性限制网络的性能。

3. 利用这些界限共同设计网络的通信拓扑结构和隐私参数。接下来，我们将定义代理和网络层面的动态特性，并详细说明每个代理如何实施差分隐私。

3.2 多智能体编队控制

编队控制的目标是使智能体达到一组相对状态。多智能体编队控制是一个研究较为深入的问题，已有多种数学模型可用于实现类似的结果[27, 28, 29, 30, 31, 32, 33]。我们将定义相互通信的智能体之间的偏移量，目标是使所有智能体保持这些相对偏移量。

假设 2 ([33] 的第 6.1.1 节)。由 $\{\Delta\}$ 指定的排列是可行的，即存在一个点集，同时满足所有成对距离。我们要求对于所有 $(i, j) \in E$ ， $\Delta = -\Delta$ 。

目标是拥有极限 $\lim_{k \rightarrow \infty}$ 对于所有 $(i, j) \in E$ ， $(x(k) - x(k)) = \Delta$ 。该构造可以以……为中心。

R 中的任何点，即，我们允许阵型具有平移不变性 [33]。

注 1. 对于所有 $i \in [N]$ ，智能体 i 将执行下文将详细描述私有化形式的(1)。为了便于讨论，我们将考虑一组点 $P = \{p, \dots, p\}$ ，满足对于所有 $(i, j) \in E$ ， $p - p = \Delta$ 。这里我们强调，智能体不知道（也不需要知道）任意 $i \in [N]$ 的点 p 。我们使用 P 仅仅是为了便于讨论和分析。下面的算法 1 展示了每个智能体实际执行的步骤，这些步骤都不需要 p 。

如式 (1) 所示，我们将智能体建模为离散时间单积分器，即 $x(k+1) = x(k) + u(k) + n(k)$ ，其中 $n(k) \sim N(0, I)$ 为过程噪声， $u(k) \in R$ 为智能体 i 的输入。令向量 $p = (p, \dots, p) \in R$ 为满足形成规范的网络级状态，令 $x(k) = (x(k), \dots, x(k)) \in R$ ，令 $n(k) = (n(k), \dots, n(k)) \in R$ 。令 $\bar{x}(k) = x(k) - p$ 。那么，在网络层面分析式 (1) 等价于分析

$$\bar{x}(k+1) = (I - \gamma L(G) \otimes I) \bar{x}(k) + n(k), \quad (2)$$

其中 $\otimes$ 是克罗内克积；我们重申，(2) 仅用于分析，代理不需要知道任何 i 的 p 。

令 $\bar{x}$ 为 $\bar{x}$ 的 l 个标量元素，其中 $l \in [d]$ 。则 $\bar{x} = [\bar{x}, \dots, \bar{x}]$ 是 l 维空间中所有智能体的状态向量，而 $\bar{n} = [n, \dots, n]$ 是相应的噪声项向量。因此，分析 (2) 等价于分析协议。

$$\bar{x}(k+1) = (I - \gamma L(G)) \bar{x}(k) + \bar{n}(k) \quad (3)$$

同时对所有 $l \in [d]$ 。

3.3 私人通信（问题 1.1 的解答）

为了私密地实现协议 (1)，智能体 j 首先选择隐私参数 $\epsilon > 0$ 、 $\delta \in (0, 1)$ 以及邻接关系 Adj ，其中 $b > 0$ 。然后，智能体 j 使用高斯机制对其状态轨迹 x 进行隐私化。令 $\tilde{x}$ 表示 x 的差分隐私版本，其中 $\tilde{x}(k) = x(k) + v(k)$ ， $v(k) \sim N(0, \Sigma)$ 。这里， $\Sigma = \sigma I$ 且 $\sigma \geq$

钾离子 $\frac{q}{K+2\epsilon}$ 引理 1 表明，这种设置

智能体 j 的状态轨迹 (ϵ, δ) 具有差分隐私性。然后，智能体 j 与其邻居共享 $\tilde{x}(k)$ 。

Thus, agent i only has access to $\tilde{x}_j(k)$ for $j \in N_i$. Plugging this into the node-level formation control protocol in (1) gives

$$\tilde{x}_i(k+1) = \tilde{x}_i(k) + \gamma \sum_{j \in N_i} w_{ij} (\tilde{x}_j(k) - \tilde{x}_i(k) - \Delta_{ij}) + n_i(k). \quad (4)$$

It is well known that (3) and (4) are mathematically equivalent (see, e.g., [33]). We analyze (3) because it is simpler, and all of these analyses apply verbatim to (4).

We present the full private formation controller in Algorithm 1, which solves Problem 1.1.

Algorithm 1: Private formation controller (Solution to Problem 1.1)

Inputs: $\epsilon_i > 0$, $\delta_i \in (0, 1/2)$, $b_i > 0$ for all $i \in [N]$, Δ_{ij} for all $(i, j) \in E$

Outputs: Privately assembled formation

for $k = 1, 2, \dots$ **do**

for $i \in [N]$ **do**

 Agent i sends $\tilde{x}_i(k)$ to agents $j \in N_i$

 Agent i receives $\tilde{x}_j(k)$ for all $j \in N_i$

 Agent i executes the state update in (4)

We note that the Gaussian input perturbation mechanism employed here is standard. The novelty of this work lies not in developing a new mechanism but instead in the system-level analysis of privacy's impact on multi-agent control performance, as well as the incorporation of this analysis into a computational framework for privacy/network co-design, which we develop across this section and Sections 4 and 5. To solve Problems 1.2 and 1.3, we will analyze this protocol at the network level.

Lemma 2. *Consider Algorithm 1 and let Assumption 1 hold. Then the network-level dynamics for each $l \in \{1, \dots, d\}$ are*

$$\bar{x}_{[l]}(k+1) = (I_N - \gamma L(\mathcal{G}))\bar{x}_{[l]}(k) + z_{[l]}(k),$$

where $z_{[l]}(k) \sim \mathcal{N}(0, \Xi_z)$, $\Xi_z = \gamma^2 A(\mathcal{G})\Xi_v A(\mathcal{G}) + \Xi_n$, $\Xi_v = \text{diag}(\sigma_1^2, \dots, \sigma_N^2)$, and $\Xi_n = \text{diag}(s_1^2, \dots, s_N^2)$.

Proof. See A.1. □

To analyze network-level performance, we define $\beta_{[l]}(k) := \frac{1}{N} \mathbf{1}^T \bar{x}_{[l]}(k) \mathbf{1}$, which is the state vector that the protocol in (3) would converge to with initial state $x_{[l]}(k)$ and without privacy or process noise. Also let

$$e_{[l]}(k) = \bar{x}_{[l]}(k) - \beta_{[l]}(k), \quad (5)$$

which is the offset of the current state from the state the protocol would converge to without noise.

4 Performance of Differentially Private Formation Control

In this section we solve Problem 1.2. First, we compute the total mean square error in a formation. Then, we use these results to derive performance bounds. In the forthcoming proofs, we will analyze each dimension separately for simplicity. However, results will be stated in terms of all dimensions simultaneously because it is that form of network-level performance that is ultimately of interest.

We quantify steady-state performance using the total mean-square error of the network, denoted e_{ss} . Agent i 's private formation control protocol in Algorithm 1 runs in $\mathbb{R}^d$, and each agent in the network runs this protocol. This is equivalent to running d identical copies of (3), which is in $\mathbb{R}^N$. Thus, using (3), we can compute the mean square error in dimension l and then multiply by d to compute e_{ss} . The mean-square error in dimension l is equal to $\lim_{k \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N E[e_{[l],i}^2(k)]$, where $e_{[l],i}(k)$ is the i^{th} element of $e_{[l]}(k)$ in (5).

Then we have $e_{ss} := \lim_{k \rightarrow \infty} \frac{d}{N} \sum_{i=1}^N E[e_{[l],i}^2(k)]$.

因此，对于 $j \in N$ ，代理 i 只能访问 $\tilde{x}(k)$ 。将此代入 (1) 中的节点级编队控制协议，得到

$$\tilde{x}(k+1) = \tilde{x}(k) + \gamma \sum_{j \in N} w_{ij} (\tilde{x}(k) - \tilde{x}(k) - \Delta) + n(k)。 \quad (4)$$

众所周知，(3) 和 (4) 在数学上是等价的（参见例如 [33]）。我们分析 (3) 是因为它更简单，而且所有这些分析都完全适用于 (4)。

我们在算法 1 中提出了完整的私有编队控制器，解决了问题 1.1。

算法 1: 私有编队控制器（问题 1.1 的解） 输入： $\epsilon > 0$, $\delta \in (0, 1/2)$ ，对于所有 $i \in [N]$, $b > 0$ ，对于所有 $(i, j) \in E$ 输出：私有编队

```

对于  $k = 1, 2, \dots$  执行
    对于  $i \in [N]$  执行
        代理  $i$  向代理  $j \in N$  发送  $\tilde{x}(k)$  代理  $i$  接收所有
         $j \in N$  的  $\tilde{x}(k)$  代理  $i$  执行 (4) 中的状态更新

```

我们注意到，这里采用的高斯输入扰动机制是标准的。这项工作的创新之处不在于开发一种新的机制，而在于对隐私对多智能体控制性能的影响进行系统级分析，并将此分析融入到隐私/网络协同设计的计算框架中，该框架将在本节以及第 4 节和第 5 节中逐步构建。为了解决问题 1.2 和 1.3，我们将从网络层面分析该协议。

引理 2. 考虑算法 1 并假设 1 成立。则每个网络的动态特性如下：

$l \in \{1, \dots, d\}$ 是

$$\bar{x}(k+1) = (I - \gamma L(G)) \bar{x}(k) + z(k),$$

其中 $z(k) \sim N(0, \Xi)$, $\Xi = \gamma A(G) \Xi A(G) + \Xi$, $\Xi = \text{diag}(\sigma, \dots, \sigma)$ ，且 $\Xi = \text{diag}(s, \dots, s)$ 。

证明。见 A.1。 □

为了分析网络级性能，我们定义 $\beta(k) := \frac{1}{d} \mathbf{1}^T \bar{x}(k)$ ，它是协议 (3) 在初始状态 $x(k)$ 且不考虑隐私或过程噪声的情况下收敛到的状态向量。此外，令

$$e(k) = \bar{x}(k) - \beta(k), \quad (5)$$

这是当前状态与协议在没有噪声的情况下将收敛到的状态之间的偏移量。

4 差异化私有形成控制的性能

本节我们将解决问题 1.2。首先，我们计算编队中的总均方误差。然后，我们利用这些结果推导出性能界限。为了简化起见，在接下来的证明中，我们将分别分析每个维度。然而，最终结果将以所有维度同时呈现，因为我们最终关注的是网络层面的性能。

我们使用网络的总均方误差来量化稳态性能，记为 e 。

算法 1 中代理 i 的私有编队控制协议在 R 中运行，网络中的每个代理都运行此协议。这等价于运行 d 个相同的 (3) 副本，该 (3) 也在 R 中运行。因此，利用 (3)，我们可以计算维度 l 的均方误差，然后乘以 d 来计算 e 。维度 l 的均方误差等于

$$\lim_{P \rightarrow \infty} \frac{1}{P} \sum_{i=1}^P \mathbb{E} \left[\frac{1}{h} \sum_{l=1}^h \mathbb{E} [e(k)]^2 \right], \text{ 其中 } e(k) \text{ 是 (5) 中 } e(k) \text{ 的元素。}$$

那么我们有 $e := \lim_{P \rightarrow \infty} \frac{1}{P} \sum_{i=1}^P \mathbb{E} [e(k)]^2$ 。

4.1 Connections with the Lyapunov Equation

The main error bound in this paper uses the fact that we can represent the total error in the system as the trace of a covariance matrix. We define $\Xi_{e[l]}(k) = E[e[l](k)e[l](k)^T]$ and $\Xi_\infty = \lim_{k \rightarrow \infty} \Xi_{e[l]}(k)$; we omit the subscript l since Ξ_∞ is the same in each dimension.

Then we have $e_{ss} = \frac{d}{N} \text{Tr}(\Xi_\infty)$. Now we will analyze the dynamics of $e[l](k)$ and $\Xi_{e[l]}(k)$. For a given $\gamma > 0$ and a given graph $\mathcal{G}$, let $\mathcal{M} = I_N - \gamma L(\mathcal{G}) - \frac{1}{N} \mathbb{1}\mathbb{1}^T$. Then we have the following.

Lemma 3. *Consider Algorithm 1, and let Assumptions 1 and 2 hold. Then the network level error $e[l](k)$ in (5) evolves via*

$$e[l](k+1) = \mathcal{M}e[l](k) + (I_N - \frac{1}{N} \mathbb{1}\mathbb{1}^T)z[l](k), \quad (6)$$

$\Xi_{e[l]}(k)$ evolves according to

$$\Xi_{e[l]}(k+1) = \mathcal{M}\Xi_{e[l]}(k)\mathcal{M} + (I_N - \frac{1}{N} \mathbb{1}\mathbb{1}^T)\Xi_z(I_N - \frac{1}{N} \mathbb{1}\mathbb{1}^T), \quad (7)$$

and $\Xi_{e[l]}(k)$ can be computed via

$$\Xi_{e[l]}(k) = \sum_{i=0}^{k-1} \mathcal{M}^i (I_N - \frac{1}{N} \mathbb{1}\mathbb{1}^T) \Xi_z (I_N - \frac{1}{N} \mathbb{1}\mathbb{1}^T) \mathcal{M}^i. \quad (8)$$

Proof. See A.2. □

The following result computes the spectral radius of a matrix that will be used in our analysis below. A similar bound is in [34], but that work selects a vector-valued gain, while we only vary γ , which is a scalar-valued stepsize.

Lemma 4. *Let N agents communicate over a graph that satisfies Assumption 1 with Laplacian $L(\mathcal{G})$. Then the maximum singular value of $\mathcal{M} = I_N - \gamma L(\mathcal{G}) - \frac{1}{N} \mathbb{1}\mathbb{1}^T$ is $\sigma_{\max}(\mathcal{M}) = 1 - \gamma \lambda_2(L)$.*

Proof. See A.3. □

We prove the existence of the steady-state error covariance matrix Ξ_∞ by showing that it solves a discrete-time Lyapunov equation that has a unique solution.

Theorem 1. *Consider N agents executing Algorithm 1, and let Assumptions 1 and 2 hold. Then Ξ_∞ is the unique solution to the discrete time Lyapunov equation*

$$\Xi_\infty = (I_N - \frac{1}{N} \mathbb{1}\mathbb{1}^T) \Xi_z (I_N - \frac{1}{N} \mathbb{1}\mathbb{1}^T) + \mathcal{M} \Xi_\infty \mathcal{M}, \quad (9)$$

where we have $\Xi_z = \gamma^2 A(\mathcal{G}) \text{diag}(\sigma_1^2, \dots, \sigma_N^2) A(\mathcal{G}) + \Xi_n$ and $\sigma_i^2 = \frac{b_i}{2\epsilon_i} (K_{\delta_i} + \sqrt{K_{\delta_i}^2 + 2\epsilon_i})$ is the variance of privacy noise.

Proof. From Lemma 3, we have $\Xi_\infty = \sum_{i=0}^{\infty} \mathcal{M}^i (I_N - \frac{1}{N} \mathbb{1}\mathbb{1}^T) \Xi_z (I_N - \frac{1}{N} \mathbb{1}\mathbb{1}^T) \mathcal{M}^i$. Let $Q = (I_N - \frac{1}{N} \mathbb{1}\mathbb{1}^T) \Xi_z (I_N - \frac{1}{N} \mathbb{1}\mathbb{1}^T)$. Using $\mathcal{M}^0 = I_N$ and factoring gives $\Xi_\infty = Q + \mathcal{M} (\sum_{i=1}^{\infty} \mathcal{M}^{i-1} Q \mathcal{M}^{i-1}) \mathcal{M} = Q + \mathcal{M} \Xi_\infty \mathcal{M}$. By Lemma 4, $\mathcal{M}$ has eigenvalues strictly less than 1, and by Prop. 2.1 in [35], (9) has a unique, symmetric solution. □

Theorem 1, along with the fact that $e_{ss} = \frac{d}{N} \text{Tr}(\Xi_\infty)$, allows us to solve for e_{ss} . Thus, given a graph $\mathcal{G}$ and privacy parameters $\{(\epsilon_i, \delta_i)\}_{i=1}^N$, we can determine the performance of the network, encoded by e_{ss} before runtime. We will use its value to design a communication topology that allows agents to be as private as possible while satisfying $e_{ss} \leq e_R$, where e_R is the maximum allowable error at steady-state.

4.1 与李雅普诺夫方程的联系

本文的主要误差界利用了系统总误差可以表示为协方差矩阵的迹这一事实。我们定义 $\Xi(k) = E[e(k)e(k)^T]$ 且 $\Xi = \lim_{k \rightarrow \infty} \Xi(k)$ ；我们省略下标 l 是因为 Ξ 在每个维度上都相同。

那么我们有 $e = \text{Tr}(\Xi)$ 。现在我们将分析 $e(k)$ 和 $\Xi(k)$ 的动态特性。对于给定的 $\gamma > 0$ 和给定的图 G ，令 $M = I - \gamma L(G) - \frac{1}{N} \mathbf{1}\mathbf{1}^T$ 。那么我们有以下结论。

引理 3. 考虑算法 1，并假设 1 和 2 成立。则式 (5) 中的网络级误差 $e(k)$ 演化如下：

$$e(k+1) = Me(k) + (I - \frac{1}{N} \mathbf{1}\mathbf{1}^T)z(k), \quad (6)$$

$\Xi(k)$ 根据以下规律演化

$$\Xi(k+1) = M\Xi(k)M + (I - \frac{1}{N} \mathbf{1}\mathbf{1}^T)\Xi(I - \frac{1}{N} \mathbf{1}\mathbf{1}^T), \quad (7)$$

$\Xi(k)$ 可以通过以下方式计算：

$$\Xi(k) = \sum_{i=0}^{k-1} M(I - \frac{1}{N} \mathbf{1}\mathbf{1}^T)\Xi(I - \frac{1}{N} \mathbf{1}\mathbf{1}^T)M. \quad (8)$$

证明。见 A.2。 □

以下结果计算了将在下文分析中使用的矩阵的谱半径。

[34] 中有一个类似的界限，但该工作选择了一个向量值增益，而我们只改变 γ ，这是一个标量值步长。

引理 4. 设 N 个智能体在一个满足假设 1 的图上进行通信，该图的拉普拉斯算子为 $L(G)$ 。则 $M = I - \gamma L(G) - \frac{1}{N} \mathbf{1}\mathbf{1}^T$ 的最大奇异值为 $\sigma(M) = 1 - \gamma\lambda(L)$ 。

证明。见 A.3。 □

我们通过证明稳态误差协方差矩阵 Ξ 满足离散时间 Lyapunov 方程且该方程有唯一解，来证明稳态误差协方差矩阵 Ξ 的存在性。

定理 1. 考虑 N 个智能体执行算法 1，并假设 1 和 2 成立。则 Ξ 是离散时间 Lyapunov 方程的唯一解。

$$\Xi = (I - \frac{1}{N} \mathbf{1}\mathbf{1}^T)\Xi(I - \frac{1}{N} \mathbf{1}\mathbf{1}^T) + M\Xi M, \quad (9)$$

其中 $\Xi = \gamma A(G) \text{diag}(\sigma, \dots, \sigma) A(G) + \Xi$ 且 $\sigma = (K + \frac{q}{K+2\epsilon})$ 是方差隐私噪音。

证明。由引理 3 可知 $\Xi = \sum_{i=0}^{P-1} M(I - \frac{1}{N} \mathbf{1}\mathbf{1}^T)\Xi(I - \frac{1}{N} \mathbf{1}\mathbf{1}^T)M$ 。令 $Q = (I - \frac{1}{N} \mathbf{1}\mathbf{1}^T)\Xi(I - \frac{1}{N} \mathbf{1}\mathbf{1}^T)$ 。利用 $M = \frac{1}{N} \mathbf{1}\mathbf{1}^T - \gamma L(G)$ 并因式分解 M 的特征值严格小于 1，并且根据 [35] 中的命题 2.1，(9) 有唯一的对称解。

定理 1 以及 $e = \text{Tr}(\Xi)$ 这一事实，使我们能够求解 e 。因此，给定一个图 G 和隐私参数 $\{(\epsilon, \delta)\}$ ，我们可以在运行时之前确定网络的性能，并用 e 进行编码。我们将利用 e 的值来设计一种通信拓扑结构，该结构允许代理尽可能地保持隐私，同时满足 $e \leq \bar{e}$ ，其中 $\bar{e}$ 是稳态下的最大允许误差。

4.2 Analytical Bound

Next we derive a scalar bound on e_{ss} in terms of the privacy parameters $\{(\epsilon_i, \delta_i)\}_{i=1}^N$, and properties of the graph $\mathcal{G}$.

Theorem 2. *Let all the conditions from Theorem 1 hold. With $e_{ss} := \lim_{k \rightarrow \infty} \frac{d}{N} \sum_{i=1}^N E \left[e_{[l],i}^2(k) \right]$, we have*

$$e_{ss} \leq \frac{\gamma d \sum_{i=1}^N C_{w_i, d_i} \frac{b_i}{2\epsilon_i} \left(K_{\delta_i} + \sqrt{K_{\delta_i}^2 + 2\epsilon_i} \right) + \frac{N-1}{N\gamma} \sum_{i=1}^N s_i^2}{N \lambda_2(L(\mathcal{G})) (2 - \gamma \lambda_2(L(\mathcal{G})))}, \quad (10)$$

where $C_{w_i, d_i} = \sum_{j=1}^N w_{ij}^2 - \frac{d_i^2}{N}$.

Proof. See B. □

Theorem 2 solves Problem 1.2, and we will use it below to design networks for both privacy and performance. For example, if the network must achieve $e_{ss} \leq e_R$, then this can be achieved by requiring that the bound in (10) is less than e_R . This requirement will impose joint constraints on the communication topology $\mathcal{G}$, the weights in the communication topology w_{ij} , and privacy parameters $\{(\epsilon_i, \delta_i)\}_{i=1}^N$, which we will use in co-design.

5 Privacy and Network Co-design

In this section we solve Problem 1.3. Our goal is to design a network that meets global performance requirements and agent-level privacy requirements. If some agents use strong privacy, then the high-variance privacy noise they use may make global performance poor, even if many other agents have only weak privacy requirements. This effect can also be influenced by the topology of the network, e.g., if an agent with strong privacy sends very noisy messages to other agents along heavily weighted edges. Privacy/network co-design thus requires unifying and balancing these tradeoffs.

This co-design framework is designed for agents that wish to participate in a joint co-design setup. Agent i is able to constrain the value of ϵ_i that is generated by the co-design problem, and this constraint is permitted with the understanding that agent i will use the value of ϵ_i that is computed in solving the co-design problem.

5.1 Co-Design Search Space

We consider N agents that implement private formation control and each agent has a minimum strength of privacy that it will accept. Our objective is to make agents as private as possible while meeting a global performance requirement. In many settings, each agent's neighborhood set will be fixed *a priori* based on hardware compatibility or physical location, and these neighborhoods specify an unweighted, undirected graph of edges that can be used.

We will design the privacy parameters for each agent and all edge weights for the given graph. That is, we can assign a zero or non-zero weight to each edge that is present, but we cannot assign non-zero weight to an edge that is absent. We denote the given unweighted graph of possible edges by $\mathcal{G}_0$ and denote its unweighted Laplacian by L_0 . We define $\mathcal{L}(L_0)$ as the space of all weighted graph Laplacians L such that $L_{ij} = 0$ if $L_{0,ij} = 0$. Co-design will find the optimal weighted Laplacian in $\mathcal{L}(L_0)$ and privacy parameters for each agent.

5.2 Co-Design Problem Statement

Below we derive the privacy/network co-design problem.

Problem 2 (Ideal Co-design). *Given an undirected, unweighted, simple graph $\mathcal{G}_0$ with Laplacian L_0 , a required global error bound e_R , an edge weight budget B , and a minimum level of privacy ϵ_i^{max} for agent i*

4.2 分析界限

接下来，我们根据隐私参数 $\{(\epsilon, \delta)\}$ 和图 G 的性质，推导出 e_{in} 的标量界限。

定理 2. 假设定理 1 中的所有条件都成立。令 $e := \lim_{k \rightarrow \infty} e(k)$ ，我们

有

$$e \leq \frac{\gamma d \sum_{i=1}^P C_i + \frac{q}{K+2\epsilon} + \sum_{i=1}^P \frac{1}{N C_i}}{N \lambda(L(G)) (2 - \gamma \lambda(L(G)))}, \quad (10)$$

其中 $C_i = \sum_{j=1}^P w_{ij}^2$ 。

证明。见 B。 □

定理 2 解决了问题 1.2，我们将在下文中利用它来设计兼顾隐私性和性能的网络。例如，如果网络必须满足 $e \leq e$ ，则可以通过要求式 (10) 中的界限小于 e 来实现。该要求将对通信拓扑 G 、通信拓扑中的权重 w 以及隐私参数 $\{(\epsilon, \delta)\}$ 施加联合约束，这些参数将在协同设计中使用。

5 隐私和网络协同设计

本节我们将解决问题 1.3。我们的目标是设计一个既满足全局性能要求又满足代理级隐私要求的网络。如果某些代理使用强隐私策略，那么它们使用的高方差隐私噪声可能会导致全局性能下降，即使其他许多代理只有弱隐私要求。网络拓扑结构也会影响这种影响，例如，如果一个具有强隐私策略的代理沿着权重很高的边向其他代理发送噪声很大的消息。因此，隐私/网络协同设计需要统一并平衡这些权衡取舍。

该协同设计框架是为希望参与联合协同设计方案的主体而设计的。

代理 i 能够约束由协同设计问题生成的 ϵ 值，并且允许这种约束，前提是代理 i 将使用在解决协同设计问题时计算出的 ϵ 值。

5.1 协同设计搜索空间

我们考虑 N 个实现私有编队控制的智能体，每个智能体都有一个可接受的最小隐私强度。我们的目标是在满足全局性能要求的前提下，尽可能提高智能体的隐私性。在许多情况下，每个智能体的邻域集合会基于硬件兼容性或物理位置预先确定，这些邻域构成一个可用的无权无向边图。

我们将为每个代理设计隐私参数，并为给定图的所有边赋予权重。也就是说，我们可以为每条存在的边赋予零权重或非零权重，但不能为不存在的边赋予非零权重。我们将给定的无权图（所有可能的边）记为 G ，并将其无权拉普拉斯算子记为 L 。我们将 $L(L)$ 定义为所有加权图拉普拉斯算子 L 的空间，其中 $L = 0$ 表示 $L = 0$ 。协同设计将在 $L(L)$ 中找到最优的加权拉普拉斯算子以及每个代理的隐私参数。

5.2 协同设计问题陈述

下面我们推导出隐私/网络协同设计问题。

问题 2（理想协同设计）。给定一个无向、无权、简单图 G ，其拉普拉斯算子为 L ，要求全局误差界限为 e ，边权重预算为 B ，以及代理 i 的最小隐私级别 ϵ_i 。

for all $i \in [N]$, to co-design privacy and agents' communication topology, solve

$$\begin{aligned}
& \min_{L(\mathcal{G}) \in \mathcal{L}(L_0), \{\epsilon_i\}_{i=1}^N} \sum_{i=1}^N \epsilon_i^2 - \lambda_2(L(\mathcal{G})) \\
& \text{subject to} \\
& \frac{\gamma d \sum_{i=1}^N C_{w_i, d_i} \frac{b_i}{2\epsilon_i} \left(K_{\delta_i} + \sqrt{K_{\delta_i}^2 + 2\epsilon_i} \right) + \frac{N-1}{N\gamma} \sum_{i=1}^N s_i^2}{N\lambda_2(L)(2 - \gamma\lambda_2(L))} \leq e_R \\
& \epsilon_i \leq \epsilon_i^{max} \quad \text{for all } i \text{ and } \text{Tr}(L(\mathcal{G})) \leq 2B.
\end{aligned}$$

In the objective function, the term $\sum_{i=1}^N \epsilon_i^2$, helps make each agent's privacy stronger since privacy is stronger as ϵ_i shrinks. We choose a quadratic function to penalize large privacy parameters more heavily, but this choice is not essential for our framework, and any convex function of the decision variables $\{\epsilon_i\}_{i \in [N]}$ is admissible. For the term $-\lambda_2(L(\mathcal{G}))$, minimizing this helps produce a network which is as connected as possible, i.e., maximizing the algebraic connectivity will produce a solution which promotes better network-level convergence properties. The first constraint requires that the total mean square error of the network is less than some user-defined value e_R . The second constraint requires that each agent's privacy level obeys the constraint that the agent has specified. The last constraint budgets how much edge weight can be used, specifically the total edge weight used may not exceed B .

Overall, solving Problem 2 will produce a communication topology and privacy parameters that will meet a global performance constraint while allowing agents to be as private as possible. This solves Problem 1.3. The main computational challenge in solving Problem 2 comes from the objective function and first constraint. Both depend on the algebraic connectivity of the graph $\mathcal{G}$, namely $\lambda_2(L(\mathcal{G}))$, and for N nodes the computation of $\lambda_2(L(\mathcal{G}))$ has $O(N^3)$ complexity [36]. Furthermore, Problem 2 is nonconvex due to the constraint on steady-state error, and thus one typically cannot expect to find a global solution to it. To address these challenges while maintaining the essential structure of the co-design problem, we develop a relaxed formulation in the following subsection.

5.3 Biconvex Co-Design

We introduce a biconvex relaxation of Problem 2 that has more favorable numerical properties while maintaining the structure of the original problem. We present the relaxed problem first, then discuss its derivation, biconvexity, and solution approach.

Problem 3 (Implemented Co-design). *Given the same inputs as Problem 2, solve*

$$\begin{aligned}
& \min_{L(\mathcal{G}) \in \mathcal{L}(L_0), \{\epsilon_i\}_{i=1}^N, y \in [0, N]} \sum_{i=1}^N \epsilon_i^2 - y \\
& \text{subject to} \\
& \frac{\gamma d \sum_{i=1}^N C_{w_i, d_i} \frac{b_i}{2\epsilon_i} \left(K_{\delta_i} + \sqrt{K_{\delta_i}^2 + 2\epsilon_i} \right) + \frac{N-1}{N\gamma} \sum_{i=1}^N s_i^2}{Ny(2 - \gamma y)} \leq e_R \\
& \epsilon_i \leq \epsilon_i^{max} \quad \text{for all } i \text{ and } \text{Tr}(L(\mathcal{G})) \leq 2B \\
& y \leq \lambda_2(L(\mathcal{G})).
\end{aligned} \tag{11}$$

To derive Problem 3, we decouple the nonconvex constraint involving $\lambda_2(L(\mathcal{G}))$ by introducing an auxiliary variable $y \in [0, N]$. We rewrite the error constraint from Problem 2 as

$$\frac{\gamma d}{Ne_R} \sum_{i=1}^N C_{w_i, d_i} \frac{b_i}{2\epsilon_i} \left(K_{\delta_i} + \sqrt{K_{\delta_i}^2 + 2\epsilon_i} \right) + \frac{N-1}{e_R N^2 \gamma} \sum_{i=1}^N s_i^2 \leq \lambda_2(L)(2 - \gamma\lambda_2(L(\mathcal{G}))),$$

and then split this into two separate constraints: first requiring that the left-hand side be bounded by $y(2 - \gamma y)$, and second requiring that $y \leq \lambda_2(L)$. The problem still requires careful handling due to the

对于所有 $i \in [N]$ ，确定代理 i 的最小隐私级别 ϵ_i ，以共同设计隐私和代理通信拓扑结构，求解

$$\begin{aligned}
 & \underset{L(G) \in \mathcal{L}(L), \{\epsilon\}}{\text{最小}} \quad \sum_{i=1}^N \epsilon - \lambda(L(G)) \\
 & \text{受} \dots \text{约束} \\
 & \frac{\gamma d \sum_{i=1}^N C - \sum_{i=1}^N \frac{q}{K+2\epsilon} + \sum_{i=1}^N \frac{P}{N C}}{N \lambda(L) (2 - \gamma \lambda(L))} \leq e \\
 & \epsilon \leq \epsilon_i \quad \text{对于所有 } i \text{ 和 } \text{Tr}(L(G)) \leq 2B。
 \end{aligned}$$

ϵ 有隐私增强函数中代理隐私保护，因为随着 ϵ 的减小，隐私保护会增强。我们选择二次函数来更严厉地惩罚较大的隐私参数，但对于我们的框架而言，这种选择并非必要，任何关于决策变量 $\{\epsilon\}$ 的凸函数都是可接受的。对于项 $-\lambda(L(G))$ ，最小化该项有助于生成尽可能连通的网络，即最大化代数连通性将产生一个有利于网络级收敛特性的解。第一个约束要求网络的总均方误差小于某个用户定义的值 e 。第二个约束要求每个代理的隐私级别遵守该代理指定的约束。最后一个约束限制了可以使用的边权重，具体而言，使用的总边权重不得超过 B 。

总的来说，解决问题 2 将产生一种通信拓扑结构和隐私参数，该结构和参数既能满足全局性能约束，又能使代理尽可能地保护自身隐私。这就解决了问题 1.3。解决问题 2 的主要计算挑战来自目标函数和第一个约束。两者都依赖于图 G 的代数连通性，即 λ 。

其复杂度为 $\mathcal{O}(N^3)$ [36]。此外，由于稳态误差的约束，问题 2 是非凸的，因此通常无法找到其全局解。为了在保持协同设计问题基本结构的同时应对这些挑战，我们在下一小节中提出了一种放宽的公式。

5.3 双凸协同设计

我们引入了问题 2 的双凸松弛问题，该问题在保持原问题结构的同时，具有更优的数值性质。我们首先介绍松弛后的问题，然后讨论其推导、双凸性以及求解方法。

问题 3（已实施协同设计）。给定与问题 2 相同的输入，求解

$$\begin{aligned}
 & \underset{L(G) \in \mathcal{L}(L), \{\epsilon\}, y \in [0, N]}{\text{最小}} \quad \sum_{i=1}^N \epsilon - y \\
 & \text{受} \dots \text{约束} \\
 & \frac{\gamma d \sum_{i=1}^N C - \sum_{i=1}^N \frac{q}{K+2\epsilon} + \sum_{i=1}^N \frac{P}{N C}}{N y (2 - \gamma y)} \leq e \\
 & \epsilon \leq \epsilon_i \quad \text{对于所有 } i \text{ 和 } \text{Tr}(L(G)) \leq 2B \\
 & y \leq \lambda(L(G))。
 \end{aligned} \tag{11}$$

为了推导问题 3，我们通过引入辅助变量 $y \in [0, N]$ 来解耦涉及 $\lambda(L(G))$ 的非凸约束。我们将问题 2 中的误差约束重写为

$$\frac{\gamma d}{2\epsilon} \sum_{i=1}^N C - \sum_{i=1}^N \frac{q}{K+2\epsilon} + \frac{N-1}{eN\gamma} \sum_{i=1}^N \frac{P}{N C} s \leq \lambda(L)(2 - \gamma \lambda(L(G))),$$

然后将其拆分为两个独立的约束：首先要求左侧以 $y(2 - \gamma y)$ 为界，其次要求 $y \leq \lambda(L)$ 。由于……，该问题仍然需要谨慎处理。

product of C_{w_i, d_i} and the function of ϵ_i in the first constraint. This product is what creates the biconvex structure that we analyze next.

Definition 5 (Biconvex program [37]). *Let $X \subseteq \mathbb{R}^m$ and $Y \subseteq \mathbb{R}^n$ be convex sets, and let $(x, y) \in X \times Y$. A function $f : X \times Y \rightarrow \mathbb{R}$ is biconvex if, for every fixed $y \in Y$, the function $x \mapsto f(x, y)$ is convex on X , and for every fixed $x \in X$, the function $y \mapsto f(x, y)$ is convex on Y . A constraint set $S \subseteq X \times Y$ is biconvex if, for every fixed $y \in Y$, the set $\{x \in X \mid (x, y) \in S\}$ is convex, and for every fixed $x \in X$, the set $\{y \in Y \mid (x, y) \in S\}$ is convex. An optimization problem of the form*

$$\min_{(x, y) \in X \times Y} f(x, y) \quad \text{subject to } (x, y) \in S$$

is called a biconvex program if f and S are biconvex.

This framework requires partitioning the decision variables of Problem 3 into two disjoint sets, and we partition them into $(L(\mathcal{G}_0), y)$ and $(\{\epsilon_i\}_{i=1}^N)$. The following theorem establishes that Problem 3 is biconvex.

Theorem 3. *Given an undirected, simple, weighted graph $\mathcal{G}_0$, a global error requirement e_R , an edge budget B , and a minimum level of privacy $\epsilon_i^{\max}$ for agent i for all $i \in [N]$, Problem 3 is a biconvex program with respect the variables $(L(\mathcal{G}_0), y)$ and $\{\epsilon_i\}_{i=1}^N$.*

Proof. The objective function (11) is positive definite and quadratic as a function of $\{\epsilon_i\}_{i=1}^N$ and is thus convex in the privacy parameters. The objective is linear in y . This implies that the objective is biconvex in $(L(\mathcal{G}_0), y)$ and $\{\epsilon_i\}_{i=1}^N$.

To analyze the error constraint

$$\frac{\gamma d}{N e_R} \sum_{i=1}^N C_{w_i, d_i} \frac{b_i}{2\epsilon_i} \left(K_{\delta_i} + \sqrt{K_{\delta_i}^2 + 2\epsilon_i} \right) + \frac{N-1}{e_R N^2} \sum_{i=1}^N s_i^2 - y(2 - \gamma y) \leq 0, \quad (12)$$

recall that $C_{w_i, d_i} = \sum_{j=1}^N w_{ij}^2 - \frac{d_i^2}{N}$ and $d_i = \sum_{j \mid (i, j) \in E} w_{ij}$. For $i \in [N]$, let $w_i = [w_{i1}, \dots, w_{iN}]^T \in \mathbb{R}^N$. Then

$$\begin{aligned} C_{w_i, d_i} &= \sum_{j=1}^N w_{ij}^2 - \frac{d_i^2}{N} = w_i^T w_i - \frac{1}{N} (\mathbf{1}^T w_i)^2 \\ &= w_i^T w_i - \frac{1}{N} w_i^T \mathbf{1} \mathbf{1}^T w_i = w_i^T \left(I_N - \frac{1}{N} \mathbf{1} \mathbf{1}^T \right) w_i. \end{aligned}$$

Note that $I_N - \frac{1}{N} \mathbf{1} \mathbf{1}^T \succeq 0$ and thus C_{w_i, d_i} is convex for all $i \in [N]$. It can be shown that $\frac{b_i}{2\epsilon_i} (K_{\delta_i} + \sqrt{K_{\delta_i}^2 + 2\epsilon_i})$ is a convex, nonincreasing function of ϵ_i for any $b_i > 0$ and $\delta_i \in (0, 1/2)$. Lastly, $-y(2 - \gamma y)$ is convex in y . Therefore, for fixed y and w_{ij} , the error constraint (12) is convex in $\{\epsilon_i\}_{i=1}^N$. Similarly, for fixed $\{\epsilon_i\}_{i=1}^N$, the error constraint (12) is convex in y and w_{ij} . Thus (12) is biconvex in $(L(\mathcal{G}_0), y)$ and $\{\epsilon_i\}_{i=1}^N$.

The algebraic connectivity $\lambda_2(L(\mathcal{G}))$ can be expressed as $\lambda_2(L(\mathcal{G})) = \min_{x \text{ s.t. } \|x\|=1, x^T \mathbf{1}=0} x^T L(\mathcal{G}) x$. This is an infimum over a set of linear maps and is therefore concave as a function of edge weights, and the constraint $y \leq \lambda_2(L(\mathcal{G}))$ is convex. Lastly, the constraint $\text{Tr}(L(\mathcal{G})) \leq 2B$ is a linear function of edge weights and is therefore convex. Overall, this shows that Problem 3 meets the definition of a biconvex program from Definition 5. $\square$

Having established that Problem 3 is biconvex, we now consider approaches for its solution. A global optimization algorithm for biconvex programs exists [37], but it guarantees global optimality by iteratively partitioning the feasible region and maintaining associated convex relaxations and bounds for each subregion of the partition. As the number of subregions grows, this method must store a growing collection of subproblems and bounding information, which can lead to exponential space complexity in the worst case. Instead, we employ an alternate convex search approach [38, Algorithm 4.1] that maintains a single iterate and alternates between two convex subproblems. At each iteration, only the current decision variables and fixed problem data are stored, and thus the space complexity scales as $O(|E| + N)$. Specifically, our algorithm alternates between: (1) solving for optimal edge weights $\{w_{ij}\}_{i, j \in [N]}$ and auxiliary variable y while fixing

由于第一个约束条件中存在函数 C 和 ϵ 的乘积，因此该问题仍需谨慎处理。正是这个乘积形成了我们接下来要分析的双凸结构。

定义 5 (双凸规划 [37])。设 $X \subseteq \mathbb{R}$ 和 $Y \subseteq \mathbb{R}$ 为凸集，且 $(x, y) \in X \times Y$ 。函数 $f: X \times Y \rightarrow \mathbb{R}$ 是双凸的，当且仅当对于每个固定的 $y \in Y$ ，函数 $x \mapsto f(x, y)$ 在 X 上是凸的，并且对于每个固定的 $x \in X$ ，函数 $y \mapsto f(x, y)$ 在 Y 上是凸的。约束集 $S \subseteq X \times Y$ 是双凸的，当且仅当对于每个固定的 $y \in Y$ ，集合 $\{x \in X \mid (x, y) \in S\}$ 是凸的，并且对于每个固定的 $x \in X$ ，集合 $\{y \in Y \mid (x, y) \in S\}$ 是凸的。形如以下形式的优化问题

$$\min_{(x,y) \in X \times Y} \text{函数 } f(x, y) \text{ 满足 } (x, y) \in S$$

如果 f 和 S 都是双凸函数，则称该规划问题为双凸规划问题。

该框架要求将问题 3 的决策变量划分为两个不相交的集合，我们将其划分为 $(L(G), y)$ 和 $\{\epsilon\}$ 。以下定理证明问题 3 是双凸的。

定理 3. 给定一个无向、简单、加权图 G ，一个全局误差要求 e ，一个边预算 B ，以及对于所有 $i \in [N]$ 的代理 i 的最小隐私级别 ϵ_i ，问题 3 是关于变量 $(L(G), y)$ 和 $\{\epsilon\}$ 的双凸规划问题。

证明。目标函数 (11) 关于 $\{\epsilon\}$ 是正定的二次函数，因此关于隐私参数是凸的。目标函数关于 y 是线性的。这意味着目标函数关于 $(L(G), y)$ 和 $\{\epsilon\}$ 是双凸的。

分析误差约束

$$\frac{y d}{\text{内}} \sum_{i=1}^N C - \frac{b}{2\epsilon} \sum_{i=1}^N \frac{q}{K+2\epsilon} + \frac{N-1}{\text{在}} \sum_{i=1}^N s - y(2 - \gamma y) \leq 0, \quad (12)$$

回想一下， $C = \sum_{j=1}^N w_j$ 和 $d = \sum_{j \mid (i,j) \in E} w_j$ 。对于 $i \in [N]$ ，令 $w = [w_1, \dots, w_N] \in \mathbb{R}^N$ 。则

$$\begin{aligned} C &= \sum_{j=1}^N w_j = \frac{d}{N} = \frac{1}{N} \sum_{j=1}^N w_j \\ &= \frac{1}{N} \sum_{j=1}^N w_j = \frac{1}{N} \sum_{j=1}^N w_j \end{aligned}$$

注意， $\frac{1}{N} \sum_{j=1}^N w_j \geq 0$ ，因此对于所有 $i \in [N]$ ， C 是凸的。可以证明：

对于任意 $b > 0$ 和 $\delta \in (0, 1/2)$ ， $\frac{b}{2\epsilon}$ 是 ϵ 的凸非增函数。最后， $-y(2 - \gamma y)$ 关于 y 是凸的。因此，对于固定的 y 和 w ，误差约束 (12) 关于 $\{\epsilon\}$ 是凸的。类似地，对于固定的 $\{\epsilon\}$ ，误差约束 (12) 关于 y 和 w 也是凸的。因此，(12) 关于 $(L(G), y)$ 和 $\{\epsilon\}$ 是双凸的。

代数连通性 $\lambda(L(G))$ 可以表示为 $\lambda(L(G)) = \min_x L(G)x$ 。

这是线性映射集合上的下确界，因此是边权重的凹函数，约束 $y \leq \lambda(L(G))$ 是凸的。最后，约束 $\text{Tr}(L(G)) \leq 2B$ 是边权重的线性函数，因此也是凸的。总而言之，这表明问题 3 满足定义 5 中双凸规划的定义。

既然已经确定问题 3 是双凸问题，我们现在考虑其求解方法。双凸规划的全局优化算法已存在 [37]，但它通过迭代地划分可行域并维护划分的每个子区域的相关凸松弛和边界来保证全局最优性。随着子区域数量的增长，该方法必须存储越来越多的子问题和边界信息，这在最坏情况下会导致指数级的空间复杂度。因此，我们采用了一种替代的凸搜索方法 [38, 算法 4.1]，该方法维护一个迭代过程，并在两个凸子问题之间交替进行。在每次迭代中，仅存储当前的决策变量和固定的问题数据，因此空间复杂度为 $O(|E| + N)$ 。具体来说，我们的算法交替进行以下操作：(1) 在固定的情况下求解最优边权重 $\{w\}$ 和辅助变量 y 。

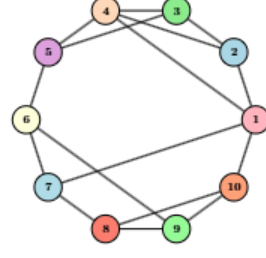


Figure 1: The graph used for simulations. Edge weights are not specified and will be optimized over.

privacy parameters $\{\epsilon_i\}_{i \in [N]}$, and (2) solving for optimal privacy parameters $\{\epsilon_i\}_{i \in [N]}$ while fixing weights and y .

This approach has several desirable properties. First, each subproblem is convex and can be solved efficiently using standard convex optimization tools. The wide availability of software for solving optimization problems therefore makes it simple to deploy this framework in practice. Second, the alternating scheme is guaranteed to converge to a stationary point since the objective function is bounded below and either decreases or remains constant at each iteration [38]. We deliberately avoid adding any more assumptions that could strengthen our convergence analysis because we want this co-design framework to have the broadest reach possible. Third, while global optimality is not analytically guaranteed, our numerical results in the following section demonstrate that our solutions yield high-quality co-designs that successfully meet both privacy and formation requirements in practice. In fact, as we will show, computational co-design results often exceed the minimum privacy requirements, exceed the minimum performance requirements, or exceed both, indicating strong levels of performance in practice. The usefulness of these stationary points follows from the analytical performance characterizations developed earlier, which ensure that any feasible stationary solution satisfies the desired privacy and steady-state performance guarantees. Strong performance numerically thus follows from the preceding analytical developments that have been made.

6 Simulations

In this section we provide simulation results for optimal privacy/network co-design¹. This section does not provide any numerical comparisons to prior work because we are not aware of any comparable prior work that co-designs a network topology and a trajectory-level differential privacy implementation. We first define an input topology as the undirected, unweighted, simple graph $\mathcal{G}$ shown in Figure 1. Then, we run privacy/network co-design to obtain edge weights and privacy parameters $\{\epsilon_i\}_{i=1}^N$. In these results, the smaller a node is drawn the more private it is, i.e., ϵ_i gets smaller as node i shrinks, and the thicker an edge is drawn the more weight it has. In simulation, edges with weights less than 10^{-4} are considered deleted. Within each example, the same solution was reached regardless of the initial iterate used by the optimization algorithm.

Example 1: (Trading off privacy and performance) Fix the input graph $\mathcal{G}$ shown in Figure 1. Fix $\epsilon^{max} = [0.4, 0.9, 0.55, 0.35, 0.8, 0.45, 0.7, 0.5, 0.52, 0.58]^T$, $B = 6$, $\gamma = 1/2N$, $\delta_i = 0.05$, and $b_i = 1$ for all i . Now let e_R take on the values $e_R \in \{8, 16, 64\}$. For each of these values, privacy/network co-design was used to design the weighted graphs shown in Figure 2 and the privacy parameters shown in Figure 3.

In Figure 2, as we allow weaker performance, quantified by larger e_R , the agents are able to use a much stronger level of privacy. This is illustrated by the nodes shrinking as e_R increases from Figure 2a to 2c. The edge weights are roughly constant due to the fixed budget of $B = 6$ and the biconvex nature of the problem. In Figure 3, we also see that as the required level of performance decreases, co-design allows the agents to be more private. This trend persists for all agents in varying magnitudes, which are influenced by the topology and each ϵ_i^{max} . Moreover, Figure 3 shows that, even under the tightest constraint on e_{ss} , agents implement values of ϵ_i ranging from approximately 0.04 to 0.07, which are in line with some of the strongest privacy protections that have been used in the literature [26]. Thus, even when using input perturbation for privacy, there is an inherent compatibility between privacy and formation control, and agents can simultaneously attain both high performance and strong privacy protections. $\triangle$

¹All code is available on GitHub [39].

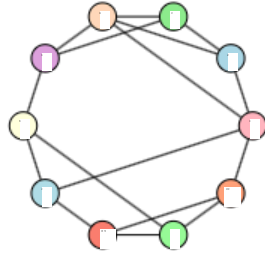


图 1: 用于模拟的图。边的权重未指定，将进行优化。

我们的算法交替进行：(1) 在固定隐私参数 $\{\epsilon\}$ 的情况下求解最优边权重 $\{w\}$ 和辅助变量 y ，以及 (2) 在固定权重和 y 的情况下求解最优隐私参数 $\{\epsilon\}$ 。

这种方法具有几个理想的特性。首先，每个子问题都是凸问题，可以使用标准的凸优化工具高效求解。因此，由于优化问题求解软件的广泛可用性，该框架在实践中易于部署。其次，由于目标函数有下界，并且在每次迭代中要么减小要么保持不变[38]，因此交替方案保证收敛到驻点。我们特意避免添加任何可能加强收敛性分析的假设，因为我们希望该协同设计框架具有尽可能广泛的适用性。第三，虽然全局最优性无法从解析上保证，但我们在下一节中的数值结果表明，我们的解决方案能够产生高质量的协同设计，并在实践中成功满足隐私和信息收集要求。事实上，正如我们将要展示的，计算协同设计结果通常超过最低隐私要求、超过最低性能要求，或两者都超过，这表明其在实践中具有很高的性能水平。这些驻点的有效性源于先前建立的分析性能表征，这些表征确保任何可行的驻点解都能满足所需的隐私性和稳态性能保证。因此，数值上的优异性能也源于先前的分析成果。

6 模拟

本节提供最优隐私/网络协同设计的仿真结果。由于我们尚未发现任何可比的、协同设计网络拓扑结构和轨迹级差分隐私实现的先前工作，因此本节不提供与先前工作的任何数值比较。首先，我们将输入拓扑结构定义为图 1 所示的无向、无权简单图 G 。然后，我们运行隐私/网络协同设计算法以获得边的权重和隐私参数 $\{\epsilon\}$ 。在这些结果中，节点绘制得越小，其隐私性就越高，即随着节点 i 的缩小， ϵ 值也越小；边绘制得越粗，其权重就越大。在仿真中，权重小于 10 的边被视为已删除。在每个示例中，无论优化算法使用何种初始迭代，最终都得到了相同的解。

示例 1: (权衡隐私和性能) 固定图 1 所示的输入图 G 。固定 $\epsilon = [0.4, 0.9, 0.55, 0.35, 0.8, 0.45, 0.7, 0.5, 0.52, 0.58]$ ， $B = 6$ ， $\gamma = 1/2N$ ， $\delta = 0.05$ ，且对于所有 i ， $b = 1$ 。现在令 e 取值 $e \in \{8, 16, 64\}$ 。对于每个 e 值，使用隐私/网络协同设计来设计图 2 所示的加权图和图 3 所示的隐私参数。

如图 2 所示，随着性能要求降低（以 e 值增大来量化），智能体能够使用更强的隐私保护级别。这体现在图 2a 到 2c 中，随着 e 值的增大，节点数量减少。由于预算固定为 $B = 6$ 以及问题的双凸性质，边的权重基本保持不变。在图 3 中，我们还可以看到，随着所需性能水平的降低，协同设计使得智能体能够实现更高的隐私保护级别。这一趋势在所有智能体中均存在，只是程度有所不同，其大小受拓扑结构和每个 ϵ_i 的影响。此外，图 3 显示，即使在对 e 值施加最严格的约束下，智能体实现的 ϵ 值也约为 0.04 到 0.07，这与文献 [26] 中使用的一些最强的隐私保护级别相符。因此，即使使用输入扰动来实现隐私保护，隐私和编队控制之间也存在固有的兼容性，智能体可以同时实现高性能和强大的隐私保护。△

¹ 所有代码均可在 GitHub 上找到 [39]。

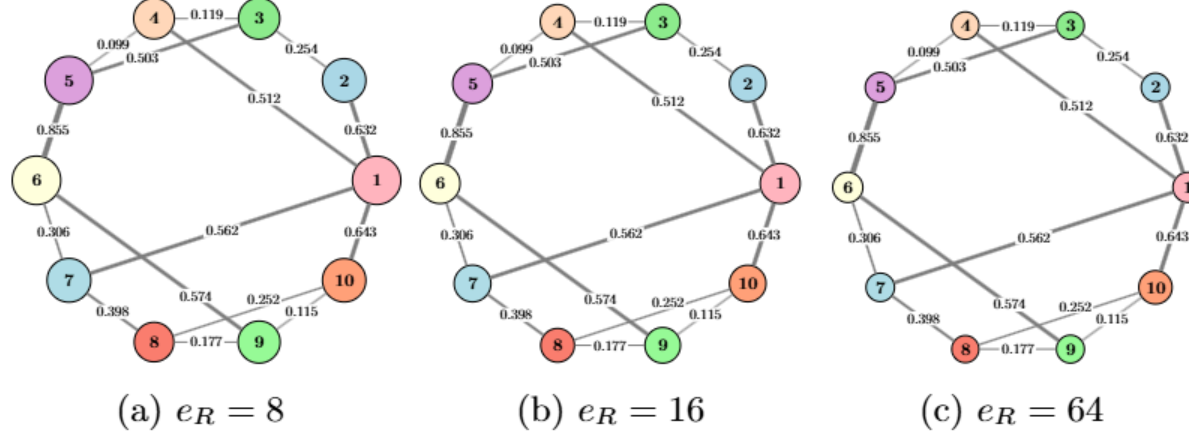


Figure 2: Graphs generated in Example 1 for $e_R \in \{8, 16, 64\}$. The smaller a node is drawn, the smaller its value of ϵ_i is. The thicker an edge is drawn, the larger its weight is. When we allow weaker performance, indicated by larger e_R , each agent has a stronger level of privacy and the topology uses less weight as illustrated by the nodes shrinking from Figure 2a to Figure 2c.



Figure 3: The privacy parameters designed in Example 1 with $e_R \in \{8, 16, 64\}$.

Example 2: (Trading off edge budget and privacy) We fix all the parameters as given in Example 1, however, we set $e_R = 16.0$ and let B take on the values $B \in \{2, 4, 8\}$, i.e., we fix a level of performance and modulate the edge weight budget. The co-designed edge weights and privacy parameters are given in Figures 4 and 5.

In Figure 4, as the budget is increased more edge weight is used and the agents get *less* private. As B is increased more edge weight is used to increase the $\lambda_2(L(\mathcal{G}))$ term of the objective. However, high edge weights amplify the privacy noise injected by agents, which has a negative impact on steady state error. Therefore, to meet the performance requirement of $e_R = 16$, co-design is making agents less private when we produce networks with high edge weights. This is directly controlled by B . The numerical values of the privacy parameters are presented in Figure 5, where we observe the privacy parameters roughly doubling between $B = 2$ and $B = 8$.

Remark 2. *These results show that the proposed privacy and network co-design framework effectively handles the increased amount of noise required by input perturbation by incorporating the privacy noise in the system analysis and design.*

7 Conclusions

We have developed a framework to design networks and privacy parameters for differentially private formation control, which allows agents to privately assemble formations with bounded steady-state error. This framework was developed for cooperative agents, and future work includes quantifying the extent to which it is robust to non-cooperative and adversarial agents. Such agents may inject false information into the network (among other attacks) and a promising direction for future work is to explore how privacy's insensitivity to changes in one agent's state trajectory can help make such attacks less effective.

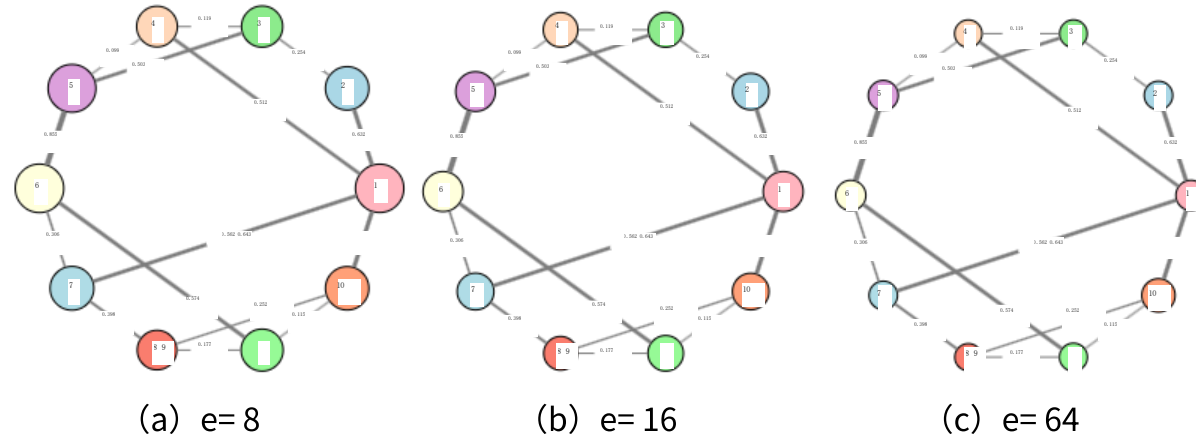


图 2: 示例 1 中 $e \in \{8, 16, 64\}$ 生成的图。节点绘制得越小，其 ϵ 值越小。边绘制得越粗，其权重越大。当我们允许性能降低（由较大的 e 值表示）时，每个代理的隐私级别更高，拓扑结构使用的权重也更少，如图 2a 到图 2c 中节点尺寸缩小所示。

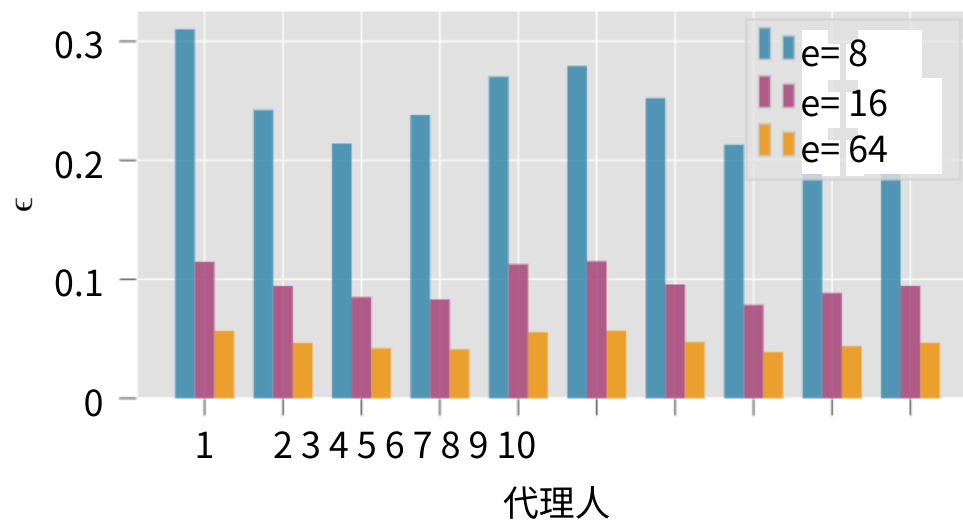


图 3: 示例 1 中设计的隐私参数，其中 $e \in \{8, 16, 64\}$ 。

例 2: (权衡边预算和隐私) 我们固定所有参数，与例 1 相同，但设定 $e = 16.0$ ，并令 B 取值于 $B \in \{2, 4, 8\}$ ，即固定性能水平并调整边权重预算。共同设计的边缘权重和隐私参数如图 4 和图 5 所示。

如图 4 所示，随着预算的增加，边的权重也随之增加，代理的隐私性降低。当 B 值增大时，边的权重增加，从而提高了目标函数中的 $\lambda(L(G))$ 项。然而，较高的边权重会放大代理注入的隐私噪声，从而对稳态误差产生负面影响。因此，为了满足 $e=16$ 的性能要求，协同设计在构建具有较高边权重的网络时会降低代理的隐私性。这直接由 B 值控制。图 5 展示了隐私参数的数值，从中我们可以观察到，隐私参数在 $B=2$ 到 $B=8$ 之间大致翻了一番。

备注 2. 这些结果表明，所提出的隐私和网络协同设计框架通过将隐私噪声纳入系统分析和设计中，有效地处理了输入扰动所需的增加的噪声量。

7 结论

我们开发了一个框架，用于设计差分隐私编队控制的网络和隐私参数，该框架允许智能体以有界稳态误差私密地组建编队。该框架最初是为合作智能体开发的，未来的工作包括量化其对非合作和对抗智能体的鲁棒性。这类智能体可能会向网络中注入虚假信息（以及其他攻击），因此，未来研究的一个有前景的方向是探索隐私对单个智能体状态轨迹变化的不敏感性如何有助于降低此类攻击的有效性。

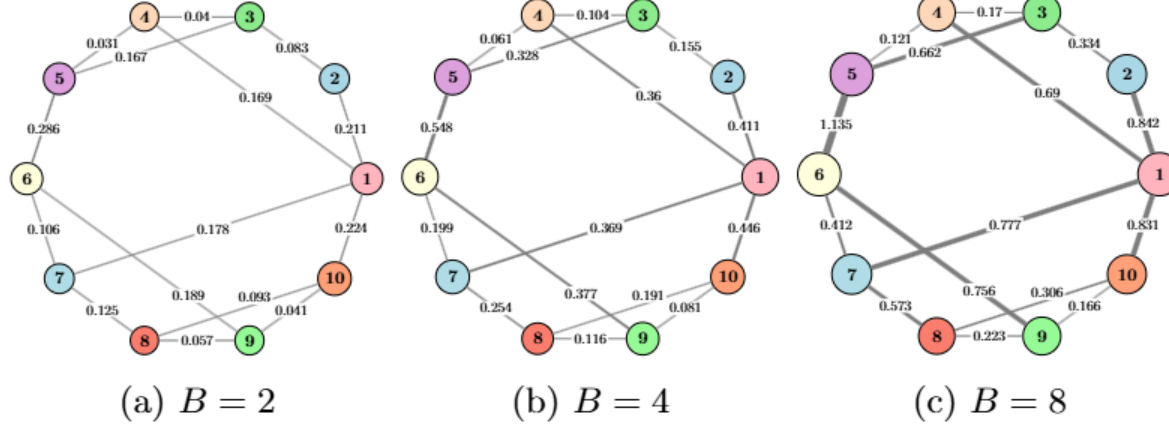


Figure 4: The outputs of privacy/network co-design with fixed parameters specified in Example 2 for different values of $B \in \{2, 4, 8\}$.

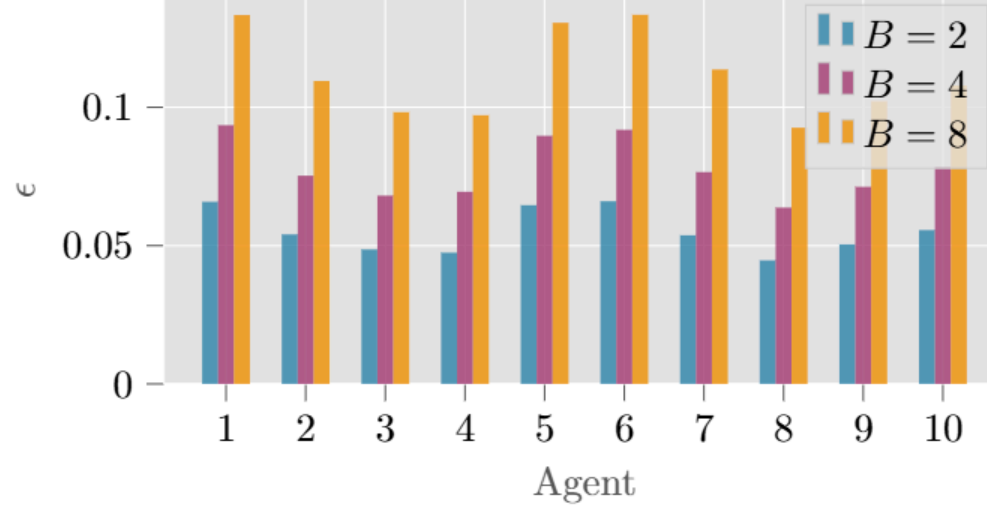


Figure 5: The privacy parameters from Example 2 with $B \in \{2, 4, 8\}$.

A Proofs of Supporting Lemmas

A.1 Proof of Lemma 2

From (4), we subtract p_i from both sides to find

$$\bar{x}_i(k+1) = \bar{x}_i(k) + \gamma \sum_{j \in N_i} w_{ij} (\bar{x}_j(k) + v_j(k) - \bar{x}_i(k)) + n_i(k),$$

which we factor as

$$\bar{x}_i(k+1) = \bar{x}_i(k) + \gamma \sum_{j \in N_i} w_{ij} (\bar{x}_j(k) - \bar{x}_i(k)) + z_i(k),$$

where $z_i(k) = \gamma \sum_{j \in N_i} w_{ij} v_j(k) + n_i(k)$. Note that $z_i(k) = \gamma ([A(\mathcal{G})]_{\text{row } i} \otimes I_d) v(k) + n_i(k)$, where $[A(\mathcal{G})]_{\text{row } i} \in \mathbb{R}^{1 \times N}$ is the i^{th} row of $A(\mathcal{G})$ and $\otimes$ is the Kronecker product. Next we define the vector

$$z_{[l]}(k) = [z_{1_{[l]}}(k)^T, \dots, z_{N_{[l]}}(k)^T]^T \in \mathbb{R}^N$$

giving $z_{[l]}(k) = \gamma A(\mathcal{G}) v_{[l]}(k) + n_{[l]}(k)$ at the network level. The covariance of $z_{[l]}(k)$ can then be calculated as $\Xi_z = E[(\gamma A(\mathcal{G}) v_{[l]}(k) + n_{[l]}(k)) (\gamma A(\mathcal{G}) v_{[l]}(k) + n_{[l]}(k))^T]$. Since $v_{[l]}(k)$ and $n_{[l]}(k)$ are statistically independent, $E[v_{[l]}(k) n_{[l]}(k)^T] = E[n_{[l]}(k) v_{[l]}(k)^T] = 0$. Applying this and the linearity of expectation gives $\Xi_z = \gamma^2 A(\mathcal{G}) \Xi_v A(\mathcal{G}) + \Xi_n$. Then $z_{[l]}(k) \sim \mathcal{N}(0, \gamma^2 A(\mathcal{G}) \Xi_v A(\mathcal{G}) + \Xi_n)$. $\square$

A.2 Proof of Lemma 3

Expanding the error term $e_{[l]}$ gives

$e_{[l]}(k+1) = (I_N - \gamma L(\mathcal{G}) - \frac{1}{N} \mathbf{1}\mathbf{1}^T) \bar{x}_{[l]}(k) + (I_N - \frac{1}{N} \mathbf{1}\mathbf{1}^T) z_{[l]}(k)$, where we have used the fact that $L(\mathcal{G}) \mathbf{1} = 0$. Let $\mathcal{M} = I_N - \gamma L(\mathcal{G}) - \frac{1}{N} \mathbf{1}\mathbf{1}^T$ and note that we also have $\mathcal{M} \beta_{[l]}(k) = (I_N - \frac{1}{N} \mathbf{1}\mathbf{1}^T) \beta_{[l]}(k) = 0$.

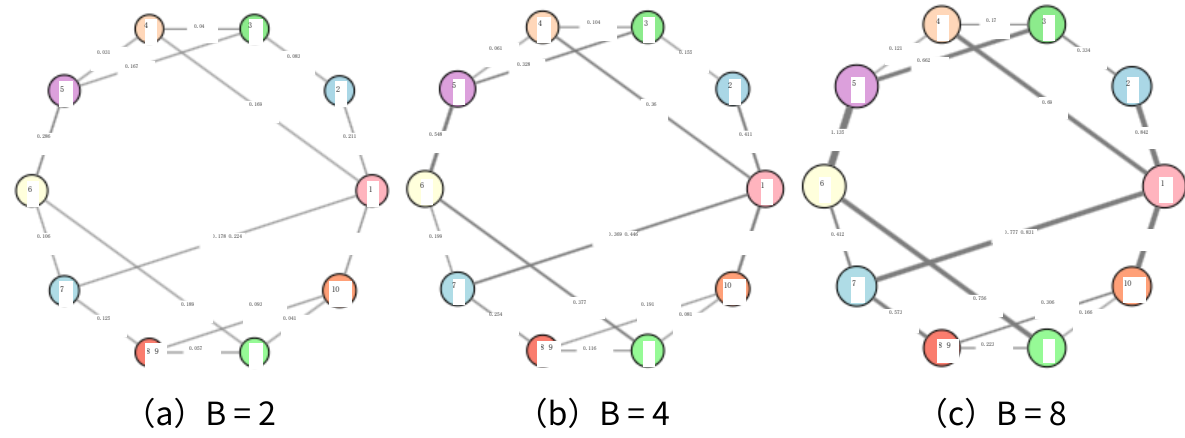


图 4: 对于示例 2 中指定的固定参数, 隐私/网络协同设计针对不同的 B 值 $\in \{2, 4, 8\}$ 的输出。

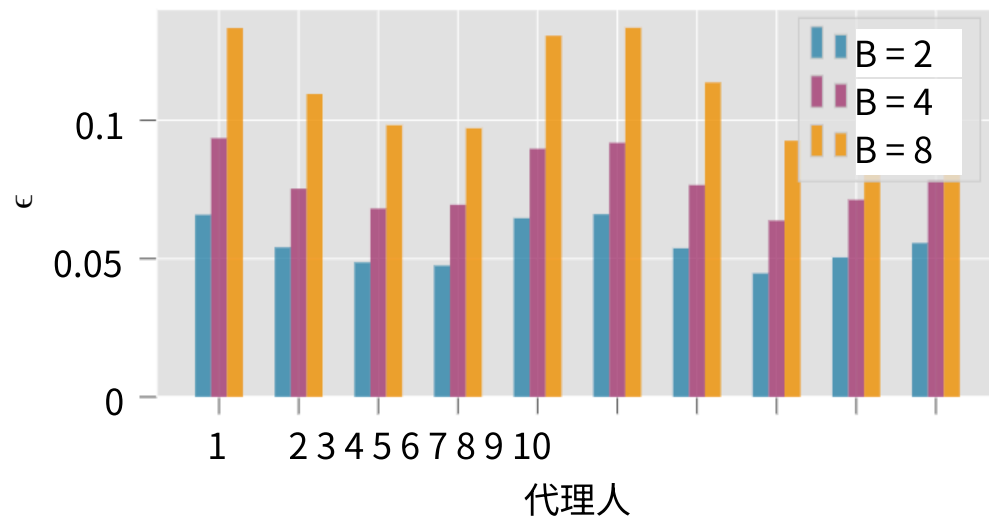


图 5: 示例 2 中 $B \in \{2, 4, 8\}$ 的隐私参数。

支持引理的证明

A.1 引理 2 的证明

由 (4) 式, 我们从等式两边减去 p , 得到

$$\bar{x}(k+1) = \bar{x}(k) + \gamma \sum_{j \in N} w_j (\bar{x}(k) + v(k) - \bar{x}(k)) + n(k),$$

我们将其考虑在内

$$\bar{x}(k+1) = \bar{x}(k) + \gamma \sum_{j \in N} w_j (\bar{x}(k) - \bar{x}(k)) + z(k),$$

其中 $z(k) = \gamma \sum_{j \in N} w_j v(k) + n(k)$ 。注意 $z(k) = \gamma A(G) v(k) + n(k)$ 的 i row, $\otimes$ 表示克罗内克积。接下来我们定义向量

$$z(k) = [z(k), \dots, z(k)] \in \mathbb{R}^P$$

在网络层面, $z(k) = \gamma A(G) v(k) + n(k)$ 。 $z(k)$ 的协方差可以计算为 $\Xi = E[z(k) z(k)^T]$

由于 $v(k)$ 和 $n(k)$ 统计独立, 因此 $E[v(k) n(k)^T] = 0$ 。应用此结论和期望的线性性质, 得到 $\Xi = \gamma A(G) \Xi A(G)^T + \Sigma$ 。则 $z(k) \sim N(0, \gamma A(G) \Xi A(G)^T + \Sigma)$ 。

□

A.2 引理 3 的证明

展开误差项得到 $e(k+1) =$

$z(k)$, 其中我们利用了 $L(G) = 0$ 这一事实。令 $x(k) = I - \gamma L(G) - 11$, 并注意到我们还有 $M\beta(k) =$

$$I - 11 - \dots$$

$$I - 11 - \dots \beta(k) = 0。$$

Then we have the equality $e_{[l]}(k+1) = \mathcal{M}e_{[l]}(k) + (I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)z_{[l]}(k)$, which proves (6). Plugging this into the definition of $\Xi_{e_{[l]}}(k+1)$ gives $\Xi_{e_{[l]}}(k+1) = \mathcal{M}\Xi_{e_{[l]}}(k)\mathcal{M} + (I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)\Xi_z(I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)$, which proves (7). Then (8) follows by applying (7) recursively. $\square$

A.3 Proof of Lemma 4

We begin by analyzing the eigenvalues of $\mathcal{M}$. Note that $\mathcal{M}\mathbb{1} = 0$, and thus $\mathcal{M}$ has eigenvalue 0 with eigenvector $\mathbb{1}$. Now let (λ, v) be an eigenpair of $\mathcal{M}$ with $\lambda \neq 0$. We will show that $\lambda \neq 0$ implies that (λ, v) is also an eigenpair of $I - \gamma L(\mathcal{G})$. Since $\mathcal{G}$ is connected and undirected, $I - \gamma L(\mathcal{G})$ is a doubly stochastic matrix. Thus, it has one eigenvalue of modulus 1 with an eigenvector of $\mathbb{1}$, and the other eigenvalues are positive and strictly inside the unit disk [32, Lemma 3].

Using $\mathbb{1}^T(I_N - \gamma L(\mathcal{G})) = \mathbb{1}^T$, we have $\mathbb{1}^T v = \mathbb{1}^T(I_N - \gamma L(\mathcal{G}))v$. Adding and subtracting $\mathbb{1}^T(\frac{1}{N}\mathbb{1}\mathbb{1}^T)v$ gives $\mathbb{1}^T v = \mathbb{1}^T \mathcal{M}v + \mathbb{1}^T(\frac{1}{N}\mathbb{1}\mathbb{1}^T)v$, then plugging in $\mathcal{M}v = \lambda v$ and $\mathbb{1}^T(\frac{1}{N}\mathbb{1}\mathbb{1}^T) = \mathbb{1}^T$ gives $\mathbb{1}^T v = \lambda \mathbb{1}^T v + \mathbb{1}^T v = (\lambda + 1)\mathbb{1}^T v$. Since $\lambda \neq 0$, we must have $\mathbb{1}^T v = 0$, and thus v is orthogonal to $\mathbb{1}$. Furthermore we have that $(\frac{1}{N}\mathbb{1}\mathbb{1}^T)v = 0$, so $\lambda v = \mathcal{M}v = (I_N - \gamma L(\mathcal{G}) - \frac{1}{N}\mathbb{1}\mathbb{1}^T)v = (I_N - \gamma L(\mathcal{G}))v$. Any non-zero eigenvalue of $\mathcal{M}$ is also an eigenvalue of $I - \gamma L(\mathcal{G})$ and the eigenvector v is orthogonal to $\mathbb{1}$.

Furthermore, $L(\mathcal{G})$ and $I_N - \gamma L(\mathcal{G})$ have the same eigenvectors and $\lambda_i(I_N - \gamma L(\mathcal{G})) = 1 - \gamma\lambda_i(L(\mathcal{G}))$. Thus, $\lambda_i(\mathcal{M}) = 1 - \gamma\lambda_i(L(\mathcal{G}))$ for $i \in \{2, \dots, N\}$, and we have $\lambda_{\max}(\mathcal{M}) = 1 - \gamma\lambda_2(L(\mathcal{G})) < 1$. Thus, all eigenvalues of $\mathcal{M}$ lie strictly in the unit disk. Now $\sigma_{\max}(\mathcal{M}) = 1 - \gamma\lambda_2(L)$ follows from the fact that $[\sigma_{\max}(\mathcal{M})]^2 = [(\lambda_{\max}((\mathcal{M})(\mathcal{M})))^{1/2}]^2 = \lambda_{\max}((\mathcal{M}))^2 = (1 - \gamma\lambda_2(L(\mathcal{G})))^2$. $\square$

B Proof of Theorem 2

With Theorem 1 and [40, Equation (151)] we have

$$Tr(\Xi_{\infty}) \leq \frac{Tr(Q)}{1 - (\sigma_{\max}(\mathcal{M}))^2}, \quad (13)$$

where $\sigma_{\max}(\mathcal{M})$ denotes the maximum singular value of $\mathcal{M}$.

We next expand $Tr(Q)$ in (13) and apply the cyclic permutation property of the trace to find

$$Tr(Q) = Tr\left((I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)(I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)\Xi_z\right). \quad (14)$$

Note that $(I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)(I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T) = I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T$. Plugging this and $\Xi_z = \gamma^2 A(G)\Xi_v A(G) + \Xi_n$ into (14) gives

$$Tr(Q) = \gamma^2 Tr\left((I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)A(G)\Xi_v A(G)\right) + Tr\left((I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)\Xi_n\right). \quad (15)$$

With $\Xi_n = \text{diag}(s_1^2, \dots, s_N^2)$, the i^{th} diagonal term of $(I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)\Xi_n$ is given by $[(I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)\Xi_n]_{ii} = (1 - \frac{1}{N})s_i^2$. Then we have that the last term in (15) is

$$Tr\left((I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)\Xi_n\right) = \frac{N-1}{N} \sum_{i=1}^N s_i^2. \quad (16)$$

To simplify $\gamma^2 Tr((I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)A(G)\Xi_v A(G))$, cyclic permutation of the trace gives that

$$\gamma^2 Tr\left((I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)A(G)\Xi_v A(G)\right) = \gamma^2 Tr\left(A(G)(I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)A(G)\Xi_v\right).$$

The i^{th} diagonal term of $A(G)(I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)A(G)$ is $[A(G)(I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)A(G)]_{ii} = \sum_{j=1}^N w_{ij}^2 - \frac{d_i^2}{N}$, and

$$\gamma^2 Tr\left(A(G)(I_N - \frac{1}{N}\mathbb{1}\mathbb{1}^T)A(G)\Xi_v\right) = \gamma^2 \sum_{i=1}^N C_{w_i, d_i} \sigma_i^2. \quad (17)$$

那么,我们证明了(6)和(7)将此代入(8)的定义,得到

$$\Xi(k+1) = M\Xi(k)M^T +$$

证明了(7)。然后,递归应用(7)即可得到(8)。

A.3 引理 4 的证明

我们首先分析矩阵 M 的特征值。注意到 $M\mathbf{1} = 0$, 因此 M 有一个特征值为 0 且特征向量为 $\mathbf{1}$ 的特征值。现在设 (λ, v) 是 M 的一个特征对, 且 $\lambda \neq 0$ 。我们将证明 $\lambda \neq 0$ 蕴含 (λ, v) 也是 $I - \gamma L(G)$ 的一个特征对。由于 G 是连通且无向的, 因此 $I - \gamma L(G)$ 是一个双随机矩阵。因此, 它有一个模为 1 且特征向量为 $\mathbf{1}$ 的特征值, 其余特征值均为正且严格位于单位圆盘内 [32, 引理 3]。

利用 $\mathbf{1}(I - \gamma L(G)) = \mathbf{1}$, 我们有 $\mathbf{1}v = \mathbf{1}(I - \gamma L(G))v$ 。加上和减去 $\mathbf{1}(11)v$ 得到 $\mathbf{1}v = \mathbf{1}Mv + \mathbf{1}(11)v$, 然后代入 $Mv = \lambda v$ 和 $\mathbf{1}(11)v = \mathbf{1}$, 得到 $\mathbf{1}v = \lambda \mathbf{1}v + \mathbf{1}v = (\lambda + 1)\mathbf{1}v$ 。由于 $\lambda \neq 0$, 我们必须有 $\mathbf{1}v = 0$, 因此 v 与 $\mathbf{1}$ 正交。此外, 我们有 $(11)v = 0$, 所以 $\lambda v = Mv = (I - \gamma L(G) - 11)v = (I - \gamma L(G))v$ 。 M 的任何非零特征值也是 $I - \gamma L(G)$ 的特征值, 并且特征向量 v 与 $\mathbf{1}$ 正交。

此外, $L(G)$ 和 $I - \gamma L(G)$ 具有相同的特征向量, 并且 $\lambda(I - \gamma L(G)) = 1 - \gamma\lambda(L(G))$ 。因此, 对于 $i \in \{2, \dots, N\}$, 有 $\lambda(M) = 1 - \gamma\lambda(L(G))$, 并且 $\lambda(M) = 1 - \gamma\lambda(L(G)) < 1$ 。因此, M 的所有特征值都严格位于单位圆盘内。现在, $\sigma(M) = 1 - \gamma\lambda(L)$ 由以下事实得出:

$$[\sigma(M)] = \lambda((M)(M))$$

B 定理 2 的证明

利用定理 1 和 [40, 公式 (151)], 我们有

$$\text{Tr}(\Xi) \leq \frac{\text{Tr}(Q)}{1 - (\sigma(M))}, \quad (13)$$

其中 $\sigma(M)$ 表示 M 的最大奇异值。

接下来, 我们将 (13) 中的 $\text{Tr}(Q)$ 展开, 并应用迹的循环置换性质来找到

$$\text{Tr}(Q) = \text{Tr} \left((I - \frac{1}{N}11)(I - \frac{1}{N}11)\Xi \right). \quad (14)$$

注意 $(I - 11)(I - 11) = I - 11$ 。将此式和 $\Xi = \gamma A(G)\Xi A(G) + \Xi$ 代入 (14) 式, 得到

$$\text{Tr}(Q) = \gamma \text{Tr} \left((I - \frac{1}{N}11)A(G)\Xi A(G) \right) + \text{Tr} \left((I - \frac{1}{N}11)\Xi \right). \quad (15)$$

令 $\Xi = \text{diag}(s, \dots, s)$, 则 $(I - 11)\Xi$ 的对角项由下式给出: $(1 - \frac{1}{N}) \cdot s$ 。那么, (15) 式中的最后一项是

$$\text{Tr} \left((I - \frac{1}{N}11)\Xi \right) = \frac{N-1}{N} \sum_{i=1}^N s_i. \quad (16)$$

为了简化 $\gamma \text{Tr} \left((I - 11)A(G)\Xi A(G) \right)$ 对迹进行循环置换, 得到

$$\gamma \text{Tr} \left((I - \frac{1}{N}11)A(G)\Xi A(G) \right) = \gamma \text{Tr} \left(A(G)(I - \frac{1}{N}11)A(G)\Xi \right).$$

$A(G)(I - 11)A(G)$ 的对角项为 $A(G)(I - 11)A(G)_{ii} = \sum_{j=1}^N w_{ij}^2$, 和

$$\gamma \text{Tr} \left(A(G)(I - \frac{1}{N}11)A(G)\Xi \right) = \gamma \sum_{i=1}^N C_i s_i. \quad (17)$$

Using (16)-(17) in (15) gives $Tr(Q) = \gamma^2 \sum_{i=1}^N C_{w_i, d_i} \sigma_i^2 + \frac{N-1}{N} \sum_{i=1}^N s_i^2$, and plugging this into (13) gives $Tr(\Xi_\infty) \leq \frac{\gamma^2 \sum_{i=1}^N C_{w_i, d_i} \sigma_i^2 + \frac{N-1}{N} \sum_{i=1}^N s_i^2}{1 - (\sigma_{\max}(\mathcal{M}))^2}$. Using Lemma 4 to expand $(\sigma_{\max}(\mathcal{M}))^2$ and then using $e_{ss} = \frac{d}{N} Tr(\Xi_\infty)$ completes the proof. $\square$

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将式 (16)-(17) 代入式 (15) 可得 $\text{Tr}(\mathbf{Q}) = \gamma \sum_{i=1}^P C\sigma + \sum_{i=1}^P s$ ，将此代入 (13) 式可得 $\text{Tr}(\Xi) \leq \frac{1}{1-(\sigma(M))} \dots$ 利用引理 4 展开 $(\sigma(M))$ ，然后利用 $e = \frac{d}{N} \text{Tr}(\Xi)$ 完成了证明。 □

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